

SPECTROSCOPIC INFRARED
ELLIPSOMETRY:
COMPONENTS, CALIBRATION,
AND APPLICATION

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Chapter 1

Introduction

Ellipsometry, introduced by Drude in 1889 [1], has the potential to become a major tool in the analysis of surfaces, interfaces, and layer systems. Among other surface analysis techniques, such as for example those applying α -particles, β -particles, X-rays, and γ -rays, ellipsometry stands out because of its non-destructive nature, its in-situ possibilities, its lack of ultra high vacuum requirements, its sensitivity, and its simple and cheap implementation. The spectroscopic infrared variant further enhances the usefulness of ellipsometry by the additionally gained spectroscopic information, which can be translated to, for example, information about electronic band structure, chemical bonds, and phonon states.

A contrast with the potential of spectroscopic infrared ellipsometry forms the actual application of this technique. Compared with ellipsometric efforts in other regions of the spectra such as the visual and ultraviolet parts, spectroscopic infrared ellipsometry has just recently started to emerge from the pioneer stages [2–16]. The reasons that ellipsometry in the infrared part of the spectrum is so late in developing are the lack of good light sources and good polarizers. A widely used light source in the infrared is the globar. Thermal light sources like the globar radiate with lower intensity in the infrared than in the visible, leaving little intensity for experiment, especially when used in combination with a grating spectrometer. However, the advent of Fourier spectrometers, introduced to ellipsometry by Röseler [17, 18], helped to overcome the intensity problem. Moreover, the developments in high density plasma arcs used as a light source [19, 20], further relieved this problem.

The lack of good polarizing components in the infrared, the other impediment for the development of spectroscopic infrared ellipsometry, is one of the main themes in this thesis. The wire grid polarizers — widely used in the infrared — only partially polarize the incident light. Similarly, imperfections in the source and detector play a role. These polarizer, source, and detector imperfections affect the results obtained with an ellipsometer and also the calibration procedure which is necessary to prepare the ellipsometer for measurements. Thus the behavior of the ellipsometer and the calibration procedure derived from it are other important items in the thesis. Of course the matter of application of spectroscopic infrared

ellipsometry is also raised.

The type of ellipsometer used for spectroscopic application up to now, has been the rotating analyzer ellipsometer. Better performance can be expected from the rotating compensator ellipsometer. However, the absence of a spectroscopic retarder with collinear incident and outgoing beams prevented realization of spectroscopic rotating compensator ellipsometers. In this thesis a spectroscopic retarder is presented that permits application as a rotating compensator.

The thesis is structured as follows:

Chapter 2 introduces spectroscopic infrared ellipsometry, the ellipsometric quantities, the Fourier spectrometer used, and a method to model layer systems. The aim of this chapter is to introduce subjects that are used throughout the thesis and to serve as a presentation of some of the basics of ellipsometry with references for further information.

Chapter 3 deals with imperfect components [5]. After an introduction to Stokes vectors and Mueller matrices, all components are investigated with respect to polarizing properties. The polarizers get special attention and the construction of a better polarizer [21], the tandem wire grid, is evaluated. The Mueller formalism is used to calculate the behavior of the complete ellipsometer, from which a calibration procedure is derived.

In Chapter 4 the spectroscopic rotating analyzer ellipsometer is applied to measure the complex refractive index of several liquids in the infrared [22, 23]. This chapter serves to demonstrate the capabilities of ellipsometry in the determination of refractive indices and does not deal with the further application of the obtained refractive index values.

A superior approach to ellipsometry, rotating compensator ellipsometry, is presented in Chapter 5. The design and characterization of the essential spectroscopic retarder is an important topic in this chapter. Furthermore, this type of ellipsometry is applied to a plasma deposited teflon layer.

Utilization of the developed diagnostic tool played an important role during the research for this PhD-thesis. In Chapter 6 an overview is given of the application of the spectroscopic ellipsometer in other fields of chemistry and physics and the direction the development of the ellipsometer is taking.

Finally, in Chapter 7 the conclusions from the preceding chapters are summarized and also several recommendations are made for future efforts in the further development of spectroscopic infrared ellipsometry.

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Chapter 2

Spectroscopic Infrared Ellipsometry

2.1 Introduction

According to Azzam and Bashara [1] ellipsometry can generally be defined as the measurement of the state of polarization of a polarized light wave, and is conducted in order to obtain information about an optical system that modifies the state of polarization. During an ellipsometric experiment polarized light is allowed to interact with the optical system under investigation. This interaction changes the state of polarization of the light wave. Measurements of the initial and final states of polarization lead to the determination of the transformation properties of the system, as described by its Jones or Mueller matrix. Further information on the optical system must then be gained by the study of the polarization modifying processes within the system, which are related to the external behavior as described by the Jones or Mueller matrix of the system.

In this definition of ellipsometry no limits are imposed on the optical system. Although there are many applications of ellipsometry in transmission experiments (transmission ellipsometry or polarimetry [2]) and scattering experiments (scattering ellipsometry [3]), the major branch in ellipsometry deals with reflection experiments [4]. Aim in these experiments is characterization of surfaces, interfaces and thin films.

The reason ellipsometry — by reputation a very difficult technique — has become more and more popular is that it may be used real-time and in-situ. It is non-perturbing, inexpensive, very sensitive, does not require ultra high vacuum conditions, and with the advance of powerful computers has become much easier to master. This holds for both the experiment and for the analysis of the results.

In this chapter the definition of the basic ellipsometric quantities followed by a brief introduction to spectroscopic ellipsometry will be given, i.e., the combination of an ellipsometer and a spectrometer setup will be presented. Furthermore, there will be an overview of the different approaches to ellipsometry. Finally there

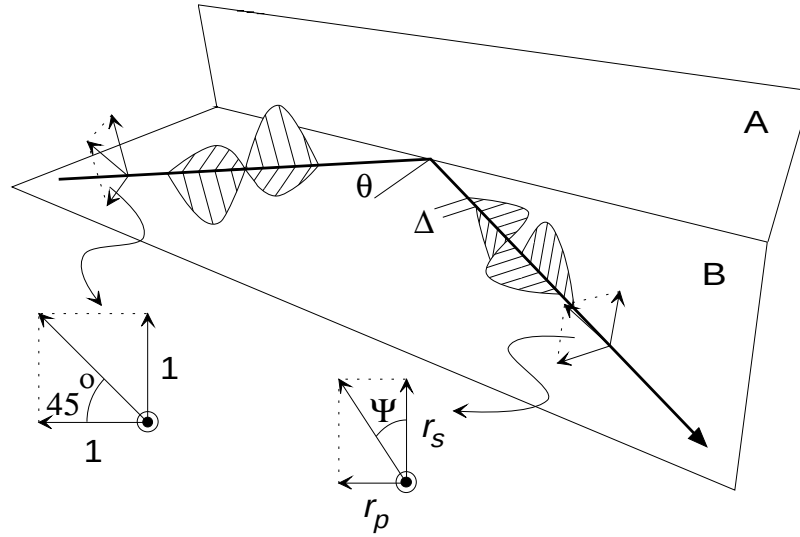


Figure 2.1: An interpretation of Ψ and Δ upon reflection of a linearly polarized beam at an interface A between two media. The plane of incidence B is determined by the incident and the reflected light beams. The p -direction is defined **p**arallel to the plane of incidence, the s -direction — **s**enkrecht — perpendicular to the plane of incidence. The incident beam has two equally large p - and s -components and is incident at an angle θ . The p -component experiences a reflection coefficient r_p , the s -component a reflection coefficient r_s .

are some sections dedicated to infrared absorption spectra, and a formalism to describe stacks of layers.

2.2 Definition of ellipsometric quantities

The ellipsometric angles Ψ and Δ originated in a time when ellipsometry still required a lot of human effort [5]. All optical components in the setup were positioned manually, and intensity was detected visually. When properly conducted, the ellipsometric quantities Ψ and Δ could directly be deduced from the angular positions of the optical components. Nowadays, ellipsometry is automated and intensity is measured photometrically. Although from the point of view of modeling ellipsometric measurement data other quantities are to be preferred over the traditional Ψ and Δ [6], they are still in use. One big advantage of the use of Ψ and Δ is their independence of the positions of the optical components, i.e., in contrast with other ellipsometric quantities such as for example the Fourier coefficients [6], Ψ and Δ are directly related to the optical parameters of the sample under investigation.

To understand this, consider the Fresnel reflection coefficients [7] and observe Fig. (2.1)

$$\begin{aligned}
r_p &= \frac{n_1 \cos \theta_0 - n_0 \cos \theta_1}{n_1 \cos \theta_0 + n_0 \cos \theta_1}, \\
r_s &= \frac{n_0 \cos \theta_0 - n_1 \cos \theta_1}{n_0 \cos \theta_0 + n_1 \cos \theta_1}.
\end{aligned}
\tag{2.1}$$

Here θ_0 represents the angle of incidence, θ_1 the angle of refraction, n_0 the refractive index of the incident or ambient medium and n_1 the refractive index of the reflecting medium. All these quantities may have complex values. The reflection coefficients may be interpreted as the eigenvalues of two eigenpolarizations of reflection. The first one, r_p , is the ratio of the E-field amplitudes after and before reflection of light with the E-field in the plane of incidence. The second, r_s , is the same ratio, but for light with the E-field perpendicular to the plane of incidence. The ratio of the two complex Fresnel reflection coefficients defines a new complex quantity ρ . This quantity ρ , a measure for the change in polarization, is usually written in polar form in terms of the two ellipsometric angles Ψ and Δ as follows

$$\rho \equiv \tan \Psi e^{j\Delta} \equiv \frac{r_p}{r_s}. \tag{2.2}$$

The tangent of the angle Ψ and the angle Δ ($0 \leq \Psi \leq 90^\circ$, $0 \leq \Delta < 360^\circ$) are the relative amplitude change and the relative phase shift between the two polarization directions, respectively.

Table 2.1: Advantages and disadvantages in several photometric ellipsometer types.

Type	Advantage	Disadvantage
RAE / RPE	medium-fast easy calibration inexpensive	inaccurate regions in Δ -range ambiguous in Δ sensitive to detector polarization (RAE) or source polarization (RPE) need for DC level
RCE	medium-fast self-calibrating non ambiguous Δ same accuracy everywhere in Ψ - Δ plane no need for DC level determination complete Stokes vector possible insensitive to source and detector polarization	extra retarder
PME	very fast insensitive to source and detector polarization	expensive ambiguous in Δ extra phase modulator

2.3 Approaches to photometric ellipsometry

Ellipsometry is usually performed by modulating the polarization state of a light beam. This is achieved by means of an optical component such as a linear polarizer — rotating analyzer (RAE [8]) and rotating polarizer ellipsometry (RPE), or a retarder — rotating compensator (RCE [9]) and phase modulation ellipsometry (PME [10]). All these approaches have some associated advantages and disadvantages, which are presented in Table (2.1). For medium-fast ellipsometry the rotating compensator method is superior to the other methods. However, in spectroscopic applications the requirement for a retarder with a well defined constant retardance over a wide spectral range poses a problem.

2.4 Spectroscopic infrared ellipsometry

In the past spectroscopic infrared ellipsometry was performed using a combination of a rotating polarizer or analyzer ellipsometer setup and a grating spectrometer [11]. Measurement of a complete spectrum required that for every wavelength in the desired spectral range a complete ellipsometric measurement had to be conducted. As a consequence, even for medium wavelength resolution, measurement times ranged from several hours to several days. This imposed heavily on the patience of the experimenter and, furthermore, ruled out any serious real-time application.

Fortunately the progress in the development of reliable Fourier transform spec-

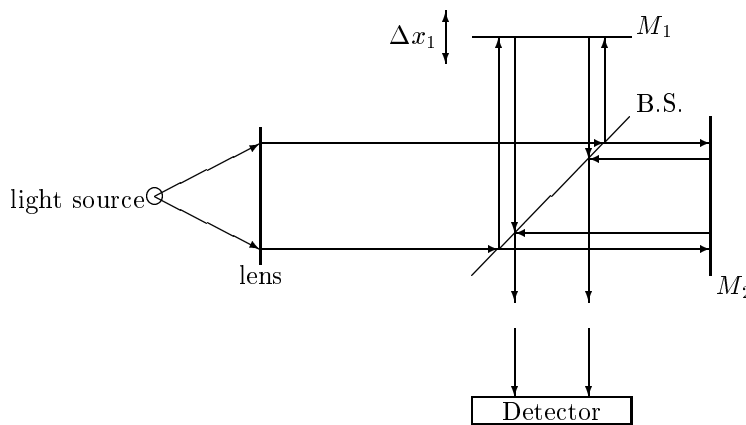


Figure 2.2: Outline of the Michelson interferometer. The light emitted by the source is collimated by the lens and divided by the beam splitter into two beams. One beam is reflected by the fixed mirror M_2 , the other beam by the moving mirror M_1 . On reaching the detector the beam coming from the moving mirror has experienced a path length difference equal to twice the relative mirror displacement Δx_1 . Depending on the wavelength of the light the beams interfere constructively or destructively.

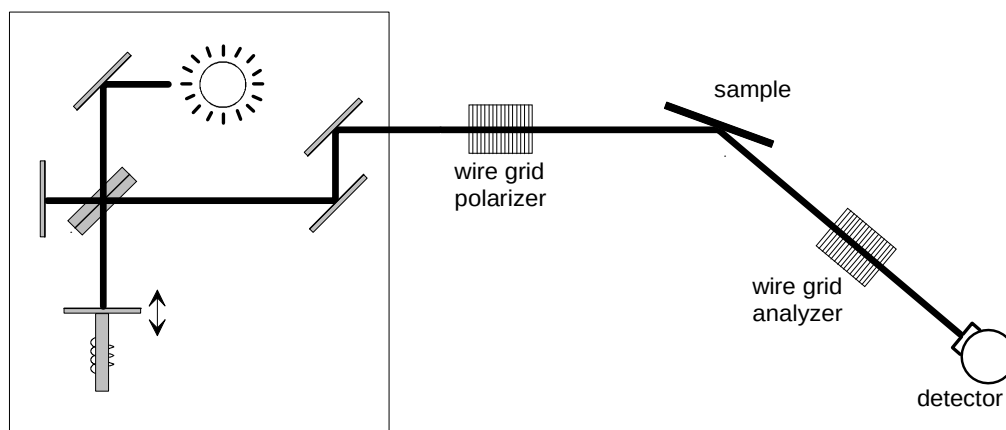


Figure 2.3: A spectroscopic Fourier transform infrared (FTIR) rotating analyzer setup. The light emitted by an internal light source (globar) passes through a Michelson interferometer and enters the ellipsometer. The first component of the ellipsometer the light passes is the polarizer, it reflects at the sample under investigation, proceeds to the analyzer, and finally hits the detector.

trometers brought relief. This type of spectrometer profits from the multiplex advantage and the large acceptance angle as compared to grating spectrometers [12, 13]. The multiplex advantage implies that all wavelengths are measured simultaneously, enhancing the signal to noise ratio. The combined advantages result in scan rates for commercially available Fourier transform spectrometers of up to 200 Hz. In combination with fast stepper motors this allows measurement times of several tens of seconds per spectrum.

A spectroscopic Fourier transform infrared (FTIR) rotating analyzer ellipsometer is a combination of a rotating analyzer setup and a Fourier transform spectrometer. The basis of the FTIR spectrometer is a Michelson interferometer shown in Fig. (2.2). This instrument modulates each wavelength by a different frequency. Fourier transformation of the subsequently detected intensity yields the desired spectrum [13]. The resulting resolution is determined by the scan length of the moving mirror, the spectral range depends on the length of the intervals between positions of the moving mirror at which the intensity is measured.

As shown in Fig. (2.3) the light leaving the Michelson interferometer enters the ellipsometer and successively passes the polarizer, sample, analyzer and finally hits the detector. In stand-alone situations both the Michelson interferometer and the rotating analyzer setup are continuously moving the mirror and the analyzer, respectively. To combine the two instruments it is necessary in the current setup to temporarily stop either of the two while the other is moving. That is, the analyzer is only moving during time intervals the Michelson interferometer is not measuring, the rest of the time the analyzer is held at a fixed position. This

means the ellipsometer has to be implemented using stepper motors to rotate the analyzer. Furthermore, the two instruments need to be synchronized, requiring some kind of communication. In a future implementation of a spectroscopic ellipsometer the absence of movement in the analyzer while the interferometer is scanning, may be dispensed with and the measurement time may be reduced. This possibility is discussed in Chapter 7.

2.5 Infrared absorption spectra

As will be shown in the next chapter, ellipsometry in the infrared region of the electromagnetic spectrum has several specific problems. Despite these problems the infrared region is very attractive, both for ellipsometry and standard spectroscopy. The main reason is the strong activity that many materials display in the infrared. In the near infrared (NIR) there are still contributions of electronic transitions [14], in the middle infrared (MIR) there are contributions of free carriers [15–17] and molecular rotation and vibration transitions [18]. Finally, in the far infrared (FIR) phonons [19] play a very important role. Most materials exhibit some activity according to either of these mechanisms, which makes infrared absorption techniques a valuable tool in the analysis of these materials.

In general the phenomena responsible for absorption in a material are described by a damped harmonic oscillator [15]. In this case the dielectric constant ϵ of the material may be considered to be the superposition of a constant background ϵ_b and one or more Lorentz profiles

$$\epsilon(\sigma) = \epsilon_b + \sum_{i=1}^k \frac{s_i \sigma_i^2}{\sigma_i^2 - \sigma^2 + j\gamma_i \sigma}. \quad (2.3)$$

Here σ is the wave number in cm^{-1} and σ_i , γ_i , and s_i represent the central wave number, width (full width half maximum), and strength of oscillator i . In total k oscillators contribute to the dielectric constant. The central wave number and width have the dimension of cm^{-1} , the oscillator strength is dimensionless. The value of the refractive index of this material is obtained by taking the square root of the dielectric constant

$$n(\sigma) = \sqrt{n_b^2 + \sum_{i=1}^k \frac{s_i \sigma_i^2}{\sigma_i^2 - \sigma^2 + j\gamma_i \sigma}}, \quad (2.4)$$

where the refractive index of the background is defined as $n_b = \sqrt{\epsilon_b}$.

The preceding description of the refractive index of a material is used in spectroscopic infrared ellipsometry to model a material by a constant background refractive index and several oscillators. Usually the model is fitted to measurement data and, subsequently, the values obtained for the oscillator central wave number, width, and strength can be used to derive the chemical bonds constituting the material and their concentrations from. The following section will

deal with a model for a stack of layer materials, into which the just presented refractive index can be introduced.

2.6 Modeling of stacked layers

A very important step in the interpretation of ellipsometric data is the simulation of ellipsometric quantities. Especially the modeling of layers on top of a substrate surrounded by an ambient plays a very important role. To describe this common experimental situation various methods have been developed in the past [20–23]. A formalism that has found widespread application is the one devised by Abelès [24–27], and implemented by McCrackin [28]. They use a 2x2 transfer matrix formalism to describe the propagation of a polarization state through a system of isotropic layers. In this approach each interface is represented by an interface matrix and the layer bulk by a propagation matrix.

A very elegant method to model a system of isotropic and even anisotropic layers is presented by Kroesen *et al.* [29]. This so called impedance formalism sets out from a substrate and a single layer and assigns impedance factors to both substrate and layer. The term *impedance formalism* is inspired by the similarities between this formalism and the one developed for the Smith chart [30]. In this latter formalism impedances in transmission lines play an important role. Using the well known reflection coefficients for a substrate with a single layer a new effective impedance factor is defined. This impedance factor thus describes the substrate-layer system as a substrate. On top of this new substrate another layer may be added or, by defining a negative thickness, a layer may be removed. In this section an adapted version of this impedance formalism will be presented, circumventing some of the problems in the original impedance formalism.

Consider the Fresnel reflection coefficients in Eqs. (2.1) for reflection at an interface between two media

$$\begin{aligned} r_p &= \frac{n_1 \cos \theta_0 - n_0 \cos \theta_1}{n_1 \cos \theta_0 + n_0 \cos \theta_1} = \frac{\frac{\cos \theta_0}{n_0} - \frac{\cos \theta_1}{n_1}}{\frac{\cos \theta_0}{n_0} + \frac{\cos \theta_1}{n_1}}, \\ r_s &= \frac{n_0 \cos \theta_0 - n_1 \cos \theta_1}{n_0 \cos \theta_0 + n_1 \cos \theta_1}. \end{aligned}$$

If the impedance factors for layer i in the p - and s -direction, f_{ip} and f_{is} , are defined by

$$\begin{aligned} f_{ip} &\equiv \frac{\cos \theta_i}{n_i} = \frac{\sqrt{n_i^2 - n_0^2 \sin^2 \theta_0}}{n_i^2}, \\ f_{is} &\equiv n_i \cos \theta_i = \sqrt{n_i^2 - n_0^2 \sin^2 \theta_0}, \end{aligned} \tag{2.5}$$

the Fresnel coefficients for the reflection at the interface of ambient and substrate

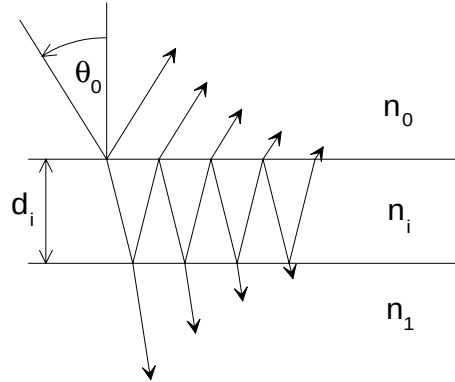


Figure 2.4: A plane electromagnetic wave propagating in an ambient with refractive index n_0 reflects at a system consisting of a substrate with refractive index n_1 covered by a layer with refractive index n_i and thickness d_i . The angle of incidence is θ_0 .

may be written in the form

$$\begin{aligned} r_p &= r_{01p} = \frac{f_{0p} - f_{1p}}{f_{0p} + f_{1p}}, \\ r_s &= r_{01s} = \frac{f_{0s} - f_{1s}}{f_{0s} + f_{1s}}. \end{aligned} \quad (2.6)$$

Note that the layer indices $i = 0$ and $i = 1$ have been reserved for the ambient and the substrate, respectively. The cosine of θ_i follows straightforwardly from Snell's law

$$n_0 \sin \theta_0 = n_i \sin \theta_i \quad \implies \quad \cos \theta_i = \sqrt{1 - \left(\frac{n_0}{n_i}\right)^2 \sin^2 \theta_0}. \quad (2.7)$$

Angle θ_i is a real number only if the angle of incidence θ_0 is below the critical angle and both refractive indices n_0 and n_i are real, i.e., both ambient and layer i are transparent. Observe the identical appearance of the expressions for the p - and s -direction. Hence the subscripts p and s may be omitted.

Now consider a plane electromagnetic wave propagating in an ambient (0) and reflecting at an angle θ_0 at a system consisting of a substrate (1) and a layer (i) depicted in Fig. (2.4). In this simple situation it is possible to calculate the effective reflection coefficient r_{0x} [31]

$$r_{0x} = \frac{r_{0i} + r_{i1} e^{-2j\beta_i}}{1 + r_{0i} r_{i1} e^{-2j\beta_i}}, \quad (2.8)$$

where β_i denotes the phase thickness defined as

$$\beta_i \equiv \frac{2\pi d_i n_i \cos \theta_i}{\lambda_0} = \frac{2\pi d_i f_{is}}{\lambda_0}. \quad (2.9)$$

Here d_i represents the thickness of the layer and λ_0 the free space wavelength. The phase thickness β_i may be interpreted as half the phase that a ray which has reflected once at the back of the layer lags a ray reflected directly at the front of the layer. According to the definitions in Eqs. (2.6) the effective reflection coefficient in Eq. (2.8) may also be written as

$$r_{0x} = \frac{f_0 - f_x}{f_0 + f_x} \equiv \frac{r_{0i} + r_{i1} e^{-2j\beta_i}}{1 + r_{0i}r_{i1} e^{-2j\beta_i}}. \quad (2.10)$$

Introducing a new quantity u by

$$u \equiv r_{i1} e^{-2j\beta_i} = \frac{f_i - f_1}{f_i + f_1} e^{-j\frac{4\pi d_i f_{is}}{\lambda_0}}, \quad (2.11)$$

Eq. (2.10) can be solved for f_x resulting in

$$f_x = f_0 \left(\frac{1 - r_{0i}}{1 + r_{0i}} \right) \left(\frac{1 - u}{1 + u} \right). \quad (2.12)$$

Again applying the definitions of Eqs. (2.6) to r_{0i} finally yields

$$f_x = f_i \left(\frac{1 - u}{1 + u} \right). \quad (2.13)$$

Thus starting with a substrate and layer for which the impedance factors are defined by Eqs. (2.5), Eq. (2.11) yields an intermediate quantity u , from which Eq. (2.13) calculates a new effective impedance factor f_x . After replacing the value of f_1 by f_x , a new layer may be added and the procedure may be repeated. Having added all layers, the effective reflection coefficient must be calculated using Eq. (2.10). To calculate the complex reflectance ratio of Eq. (2.2), the procedure just described has to be performed for both the p - and s -direction. Note that the phase thickness in Eq. (2.9) is defined in terms of the s -direction impedance factor.

As already mentioned the impedance formalism can be applied to anisotropic media, i.e., the refractive index has different values for waves traveling in the p - and s -direction. The impedance factors in Eqs. (2.5) then take a slightly different form [29]

$$\begin{aligned} f_p &\equiv \frac{\cos \theta_p}{n_s} = \frac{\sqrt{n_p^2 - n_0^2 \sin^2 \theta_0}}{n_p n_s}, \\ f_s &\equiv n_s \cos \theta_s = \sqrt{n_s^2 - n_0^2 \sin^2 \theta_0}, \end{aligned} \quad (2.14)$$

where n_p and n_s represent the refractive indices of the medium for the p - and the s -direction. Phase thickness β is redefined for both directions by

$$\begin{aligned}
\beta_p &\equiv \frac{2\pi d n_s^2}{f_p \lambda_0}, \\
\beta_s &\equiv \frac{2\pi d f_s}{\lambda_0}.
\end{aligned}
\tag{2.15}$$

2.7 Simulation and fitting of spectroscopic, ellipsometric data

To be able to simulate and fit Ψ and Δ spectra obtained in spectroscopic ellipsometry, the impedance formalism for stacks of layers from the previous section is combined with the description for the refractive index in Section (2.5). The implementation of this spectroscopic ellipsometric model in a user friendly, intuitive environment has the following main features

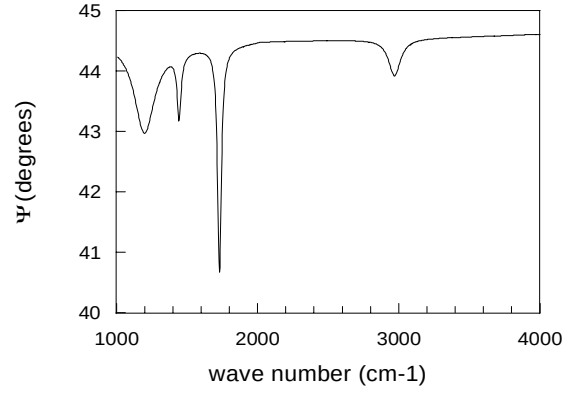
- A layer system is defined as a semi infinite substrate, an arbitrary number of layers of arbitrary thickness, and a semi infinite ambient.
- For each layer material, including substrate and ambient material, the refractive index can be defined as a fixed value, a series of values read from a file, or a series of harmonic oscillators added to a fixed background.
- Simulations and fits can be done as a function of wave number, angle of incidence, or thickness of one of the constituting layers.
- In fitting the background refractive index, the thickness, and for each oscillator the central wave number, width, and strength are determined by minimizing the sum of squares of differences between measured and calculated values of the change in polarization ρ . Any fit parameter can be fixed or made variable during this Levenberg-Marquardt minimization procedure.

An example of a spectroscopic simulation is given in Fig. (2.5), where on a gold substrate a layer of 200 nm of polymethyl-methacrylate (PMMA) is deposited.

Table 2.2: Characteristics of the harmonic oscillators in polymethyl-methacrylate. The background refractive index is $1.48 - j0.00$.

central wave number (cm^{-1})	width (FWHM) (cm^{-1})	strength
1185	175	0.0516
1441	33	0.0045
1725	32	0.0111
2968	101	0.0024

(a)



(b)

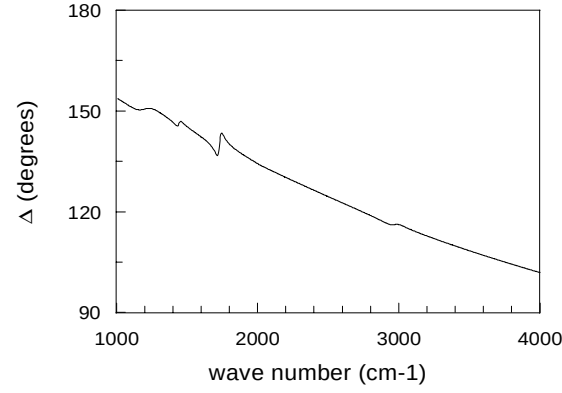


Figure 2.5: Simulated Ψ (a) and Δ (b) for a gold substrate with a 200 nm PMMA layer in an air ambient. The PMMA is characterized by a background refractive index of $1.48 - j 0.00$ and 4 harmonic oscillators. The characteristics of these oscillators are given in Table (2.2). The angle of incidence is 70° .

The ambient is air. The quantities defining the refractive index using Eq. (2.4) are given by Table (2.2).

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Chapter 3

Imperfect components in the rotating analyzer ellipsometer setup and ellipsometer calibration

3.1 Introduction

If not taken into consideration, imperfect components in a spectroscopic rotating analyzer ellipsometer may strongly affect the resulting Ψ and Δ spectra. In order to account for these errors, it is necessary to investigate all components in the setup — polarizers, source, and detector — with respect to attenuating and retarding properties. To express these properties, use is made of the Mueller formalism. Having established a description of all optical components in the setup, it is possible to calculate the exact behavior of the ellipsometer. This knowledge is imperative for obtaining reliable results with an imperfect setup. Furthermore, knowledge of this behavior is very important for the choice of a calibration procedure for spectroscopic ellipsometers. During the calibration procedure, the positions of all components relative to the plane of incidence of the sample under investigation, are determined.

This chapter starts with an introduction to Stokes vectors and Mueller matrices, and continues with a treatment of the imperfect components within the ellipsometric setup. Especially an improved polarizer — the tandem polarizer — will be discussed at some length. The chapter ends with the treatment of a calibration procedure.

3.2 Stokes vectors and Mueller matrices

A representation of the state of polarization of light was given by G.G. Stokes in 1852. He introduced four quantities — Stokes parameters [1] — which are

functions only of observables of the electromagnetic wave. The state of polarization of a beam of light, can be described in terms of these Stokes parameters. A combination of the four Stokes parameters constitutes a Stokes vector and is defined by

$$\underline{\mathbf{S}} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} I_0 \\ I_x - I_y \\ I_{+\frac{\pi}{4}} - I_{-\frac{\pi}{4}} \\ I_r - I_l \end{pmatrix}. \quad (3.1)$$

Here $S_0 \cdots S_3$ give the four Stokes parameters. Intensity is represented by the symbol I , where the subscripts indicate a direction: x and y are the horizontal and vertical directions, $+\pi/4$ and $-\pi/4$ are $+45^\circ$ and -45° azimuths, and r and l are right and left circular directions. The orientation of the horizontal axis may be fixed arbitrarily: the other directions are defined relative to this axis. Note that S_0 is simply the total intensity in the beam and S_1 , S_2 , and S_3 specify the state of polarization. Thus S_1 reflects a tendency for the polarization towards either a horizontal polarization state ($S_1 > 0$) or a vertical one ($S_1 < 0$). The parameter S_1 may be determined by placing a linear polarizer in the beam. Subtraction of intensities measured at a horizontal and a vertical position of the polarizer subsequently yields the desired Stokes parameter. Similarly, S_2 implies a tendency towards a polarization state oriented either in the $+45^\circ$ ($S_2 > 0$) or the -45° direction ($S_2 < 0$). The value of S_2 is now found by subtraction of intensities measured at $+45^\circ$ and -45° of the linear polarizer. Finally, S_3 indicates a tendency for the polarization towards either a right handed ($S_3 > 0$) or a left handed circular polarization state ($S_3 < 0$). To determine this parameter either a left or right circular polarizer is necessary. Suppose a left circular polarizer is placed in the beam and the intensity is measured. Subtraction of this intensity from the intensity measured with the polarizer fully turned around, such that the light is now incident at the backside of the polarizer, gives the desired S_3 parameter.

In order to make use of the Stokes vectors, it is necessary to have a formalism that describes the effect of optical components on this polarization state, in terms of the Stokes vectors. Mueller introduced such a formalism [2], suitably named the Mueller formalism, that describes each optical component by a 4x4 matrix. Multiplying this Mueller matrix with a Stokes vector results in a new Stokes vector, that describes the polarization state of the light beam just behind the component. This procedure can be repeated, and a complete train of optical components may be described by one Mueller matrix, which is the product of the separate component Mueller matrices. Table (3.1) presents several Mueller matrices for components which are often used in ellipsometry.

Another widely used formalism to describe the state of polarization and the transformation of this state under the influence of optical components, is the one presented by Clark Jones [5]. Instead of dealing with directly observable quantities such as the intensity, this so-called Jones formalism makes use of the electric

Table 3.1: Mueller matrices for several components that are often used in ellipsometry.

Optical component	Mueller matrix
1) A perfect linear polarizer with horizontal transmission axis.	$\frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
2) A partial linear polarizer [3] with horizontal transmission axis and attenuation coefficient α .	$\frac{1}{2} \begin{bmatrix} 1 + \alpha & 1 - \alpha & 0 & 0 \\ 1 - \alpha & 1 + \alpha & 0 & 0 \\ 0 & 0 & 2\sqrt{\alpha} & 0 \\ 0 & 0 & 0 & 2\sqrt{\alpha} \end{bmatrix}$
3) A perfect retarder with horizontal fast axis and retardance δ .	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \delta & \sin \delta \\ 0 & 0 & -\sin \delta & \cos \delta \end{bmatrix}$
4) An imperfect retarder with a horizontal fast axis, a retardance Δ_c , and a relative amplitude change Ψ_c . Observe that Ψ_c is defined differently [4] from Ψ of the sample: $\Psi = 90^\circ - \Psi_c$.	$\begin{bmatrix} 1 & \cos 2\Psi_c & 0 & 0 \\ \cos 2\Psi_c & 1 & 0 & 0 \\ 0 & 0 & \sin 2\Psi_c \cos \Delta_c & \sin 2\Psi_c \sin \Delta_c \\ 0 & 0 & -\sin 2\Psi_c \sin \Delta_c & \sin 2\Psi_c \cos \Delta_c \end{bmatrix}$
5) Ellipsometric reflection sample where the major axis is determined by the plane of incidence. Angles Ψ and Δ are the well known ellipsometric quantities.	$\begin{bmatrix} 1 & -\cos 2\Psi & 0 & 0 \\ -\cos 2\Psi & 1 & 0 & 0 \\ 0 & 0 & \sin 2\Psi \cos \Delta & \sin 2\Psi \sin \Delta \\ 0 & 0 & -\sin 2\Psi \sin \Delta & \sin 2\Psi \cos \Delta \end{bmatrix}$
6) A depolarizing component with depolarization coefficient D .	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & D & 0 & 0 \\ 0 & 0 & D & 0 \\ 0 & 0 & 0 & D \end{bmatrix}$
7) A matrix to represent a component in another frame of reference. The new frame is rotated over an angle θ relative to the old frame of reference.	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta & 0 \\ 0 & -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

field vector. This 2-element Jones vector is transformed by an optical component 2x2 Jones matrix into a similar Jones vector. Any optical component that can be represented by a Jones matrix, may also be expressed by a Mueller matrix [6, 7]. The opposite is not true: only a pure Mueller matrix [8, 9] can be transformed into a Jones matrix. Optical components represented by a pure Mueller matrix are also designated by the terms *totally polarizing* and *nondepolarizing*. Except for the depolarizing component, all optical components in Table (3.1) have pure Mueller matrices. Any combination of two of these components again gives a pure Mueller matrix.

Although the Jones formalism is applicable to coherent light, the Mueller formalism has gradually gained more and more attention. The reason is that the Mueller formalism and its accompanying Stokes vectors directly deal with observable quantities, i.e., in contrast with the Jones description, there is no need for a complicated complex conjugation at the end of a calculation of the intensity. Nor are there any complex numbers involved. Furthermore, the Mueller representation is able to deal with partially polarized light. Finally, because of the employment of computers in algebra the 4x4 Mueller matrices are as easy to implement as the 2x2 Jones matrices.

3.3 Polarizers in the infrared

One of the components of major importance in ellipsometry is the polarizer. However, in the infrared region of the spectrum no perfect polarizer is available. Thus, to work in the infrared at all, use is made of imperfect polarizers. In the widest use are Rochon, Brewster, and wire grid polarizers, each associated with specific advantages and disadvantages. Rochon polarizers, coming from the class of birefringent polarizers, have a very good attenuation coefficient ($\alpha \approx 10^{-5}$) but have a small angle of acceptance and are usable only in the near IR because most birefringent materials become opaque for wavelengths greater than about $5 \mu\text{m}$. Brewster polarizers, from the class of reflection polarizers, perform well over a wide wavelength range, but their disadvantages are that the acceptance angle is very small, the Brewster angle changes with wavelength, and the collinearity of the incident and outgoing beam is disturbed. Wire grid polarizers, from the class of dichroic polarizers, on the other hand, do not suffer these disadvantages in acceptance angle, dispersion and collinearity disturbance but perform poorly in the near IR ($10^{-2} < \alpha < 10^{-1}$).

In this section the description of a wire grid polarizer as a partial polarizer is extended to describe the retarding properties of a wire grid polarizer [10]. Measurements that characterize this imperfect component are presented. To improve the wire grid performance, the possibility of placing two wire grids in tandem is investigated [11]. The expected improvement in performance is verified experimentally.

3.3.1 Extended description and characterization of a single wire grid polarizer

A wire grid polarizer generally consists of a substrate that is transparent over a large wavelength range, covered with a grid of evenly spaced conducting wires. For wavelengths larger than the wire spacing the component of the incident light parallel to the wires is strongly absorbed, whereas the perpendicular component passes unaffected. When the wavelength decreases to the order of the wire spacing, a substantial part of the parallel component is able to pass without being absorbed. In this case the polarizer may be regarded as a partial polarizer (see Table (3.1), position 2) with an attenuation coefficient α which reflects the quality of the polarizer. The attenuation coefficient α is defined as the ratio of minimum and maximum transmittances so $\alpha=0$ and $\alpha=1$ describe a perfect polarizer and a non-polarizing component, respectively. Another widely used measure for the quality of a polarizer is the extinction ratio. This extinction ratio ϵ , defined as the ratio of the transmittance of two polarizers with transmission axes parallel and the transmittance of two polarizers with transmission axes perpendicular, is related to the attenuation coefficient by

$$\epsilon = \frac{1}{2\alpha}, \quad \text{for} \quad \alpha \ll 1. \quad (3.2)$$

Now this description of the wire grid polarizer needs to be extended because, as Stobie and Dignam [12] following Marcuvitz [13] calculated, the grid introduces a

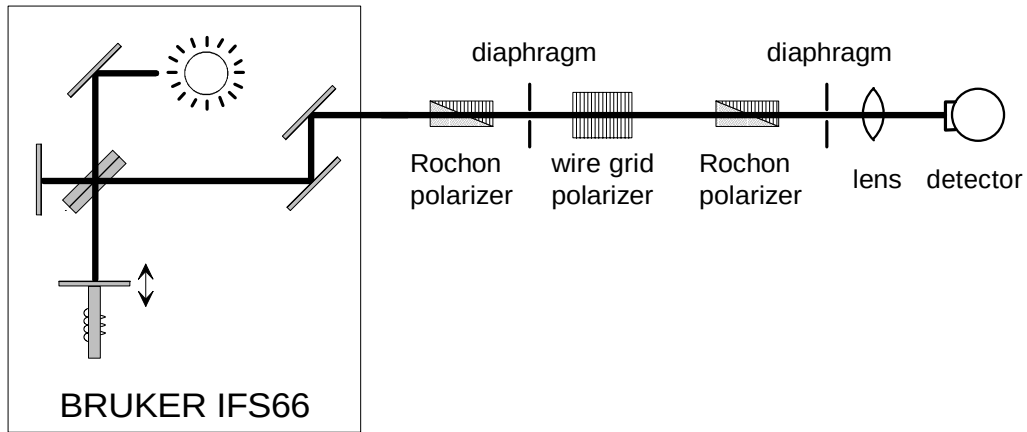
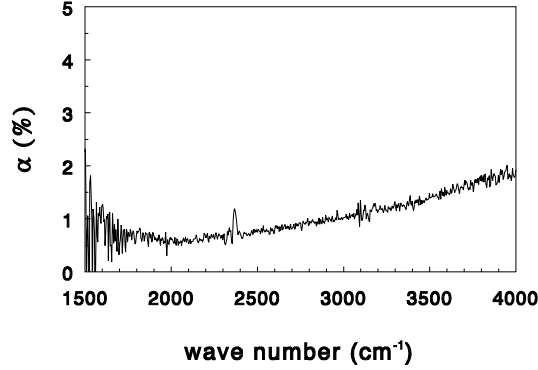


Figure 3.1: Setup to measure the characteristics of a wire grid polarizer. The first Rochon polarizer is fixed at an arbitrary position. The wire grid polarizer under investigation and the second Rochon are put at a position of 45° relative to the fixed Rochon. Starting at this position the second Rochon is rotated in steps. At each position the Bruker IFS66 spectrometer measures a spectrum. The diaphragms serve to block the extraordinary beams that also emerge from the Rochon polarizers.

(a)



(b)

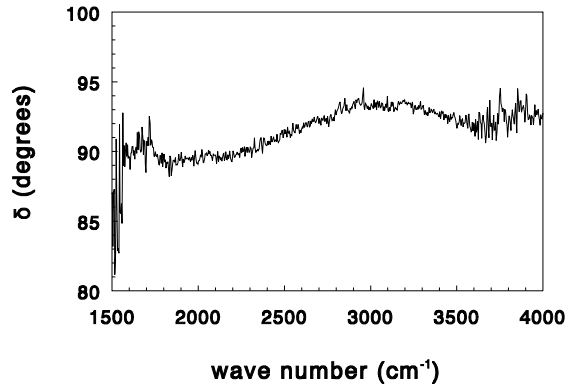


Figure 3.2: (a) Attenuation coefficient α and (b) phase difference δ between transmitted and attenuated components measured for a wire grid on a BaF_2 substrate with a wire spacing of $0.25\mu\text{m}$. The graphs show artifacts from water (1500 cm^{-1} – 1900 cm^{-1}) and CO_2 (2350 cm^{-1}) in air.

phase difference between the transmitted and attenuated components. To account for the phase difference, caused by the geometry and periodicity of the grid, the expression for the partial polarizer is multiplied by a Mueller matrix for a retarding component with a retardation δ , and the fast axis parallel to the transmission axis of the polarizer. In this case it does not matter whether the retarder is in front of or behind the polarizer. Both expressions can be found in Table (3.1) at positions 2 and 3, respectively. Multiplication leads to the matrix

$$\frac{1}{2} \begin{bmatrix} 1 + \alpha & 1 - \alpha & 0 & 0 \\ 1 - \alpha & 1 + \alpha & 0 & 0 \\ 0 & 0 & 2\sqrt{\alpha} \cos \delta & 2\sqrt{\alpha} \sin \delta \\ 0 & 0 & -2\sqrt{\alpha} \sin \delta & 2\sqrt{\alpha} \cos \delta \end{bmatrix}. \quad (3.3)$$

Stobie and Dignam showed that the attenuated component lags the transmitted component by a typical value of 90° . The exact value of the phase difference depends on the ratio of the wire thickness to the wire spacing, and the wavelength

of the incident light, but is within 10% of this typical value in any usable region of the spectrum. Deviations from 90° occur for wavelengths approaching the wire spacing.

Experimental support for the attenuation coefficient and the phase difference is gained by putting a wire grid polarizer between two MgF_2 Rochon polarizers of the class of birefringent polarizers. This setup is depicted in Fig. (3.1). The Rochon polarizers may be considered perfect up to almost $7 \mu\text{m}$ (1500 cm^{-1}) where the MgF_2 starts to become opaque and loses its birefringence. The transmitting angles relative to the first Rochon are 45° for both the wire grid and the second Rochon. With this starting point the second Rochon is rotated in steps. For each step the spectrum in the near IR is measured. From a cross-section through these spectra at a fixed wave number [14] the Fourier coefficients may be determined. Evaluation of the product of the Mueller matrices representing the components in the setup yields the relation between the obtained Fourier coefficients and the desired attenuation coefficient α and phase difference δ . Inserting the Fourier coefficients in these relations and repeating this for all wave numbers finally results in Fig. (3.2).

3.3.2 Improved polarizer: the tandem wire grid

To improve the performance of wire grids, the use of two of such polarizers in tandem is investigated. Mathematically this means multiplication of two matrices given by Eq. (3.3). Assuming that the transmission axes are parallel it is not necessary to rotate the coordinate basis of either of the polarizers. Multiplication then yields

$$\frac{1}{2} \begin{bmatrix} 1 + \alpha^2 & 1 - \alpha^2 & 0 & 0 \\ 1 - \alpha^2 & 1 + \alpha^2 & 0 & 0 \\ 0 & 0 & 2\alpha \cos 2\delta & 2\alpha \sin 2\delta \\ 0 & 0 & -2\alpha \sin 2\delta & 2\alpha \cos 2\delta \end{bmatrix}. \quad (3.4)$$

Comparison of matrix (3.3) with matrix (3.4) shows that the single polarizer and the tandem polarizer are very similar in appearance except for the attenuation coefficient α and the retardation δ : α appears to be squared, whereas δ is doubled. Thus the quality of the polarizer appears to have improved quadratically: e.g., two polarizers with attenuation coefficient 0.01 (1%) will make a tandem polarizer with attenuation coefficient 0.0001 (0.01%).

To investigate if this improvement in performance is sensitive to misalignment of the transmission axes of the two polarizers the alignment error angle ϕ is introduced. The Mueller matrix for the misaligned tandem polarizer can then be written as

$$\underline{\underline{\mathbf{M}}}_{err} = \underline{\underline{\mathbf{M}}}_{\alpha\delta} \underline{\underline{\mathbf{R}}}(-\phi) \underline{\underline{\mathbf{M}}}_{\alpha\delta} \underline{\underline{\mathbf{R}}}(\phi), \quad (3.5)$$

where $\underline{\underline{\mathbf{R}}}$ is the rotation matrix in Table (3.1) at position 7, necessary to rotate the eigencoordinate system of the first polarizer, and $\underline{\underline{\mathbf{M}}}_{\alpha\delta}$ is matrix (3.3). The resulting attenuation coefficient and transmission axis azimuth may be calculated

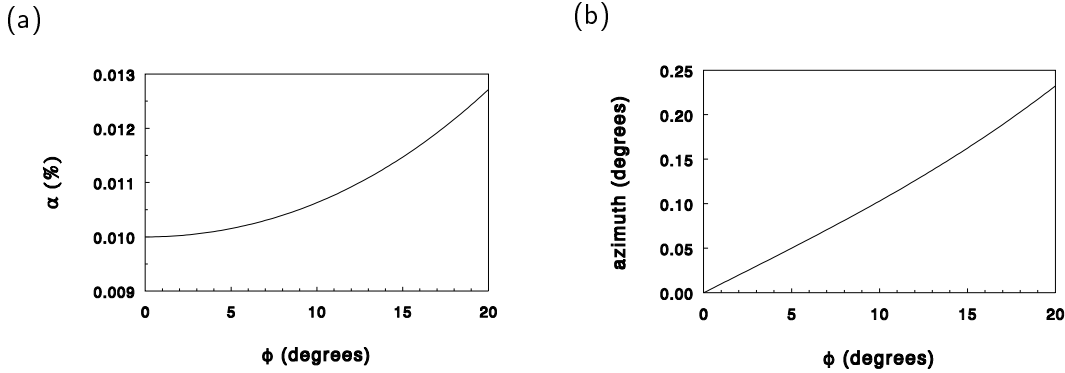


Figure 3.3: (a) Attenuation coefficient α of the tandem polarizer as a function of the error angle ϕ . (b) Azimuth of the transmission axis of the tandem polarizer as a function of the error angle ϕ . For a single polarizer, $\alpha = 1\%$ and $\delta = 90^\circ$ are assumed. The small azimuth angles relative to the second polarizer indicate that the azimuth is largely determined by the second polarizer.

by solving the eigenvalue problem [15, 16] for this misaligned tandem polarizer. The attenuation coefficient α is then the eigenvalue of the attenuated eigenvector, with the azimuth of the transmission axis deduced from the orientation of the transmitted eigenvector. The solutions of this eigenvalue problem are given in Fig. (3.3).

As might be expected on symmetry grounds, the attenuation coefficient in Fig. (3.3a) is only in second order dependent on the alignment error. The transmission azimuth in Fig. (3.3b) depends in first order on the error angle but is relatively insensitive to changes in the alignment error. Since the angles are defined relative to the second wire grid (Eq. (3.5)), this implies that the azimuth of the transmission axis of the tandem is largely determined by the azimuth of the transmission axis of the second wire grid.

An illustration to the improvement in performance is given by Fig. (3.4). The data are obtained by putting a high quality polarizer (MgF₂-Rochon, $\alpha \approx 10^{-5}$), again considered to be perfect, behind a spectrometer. The wire grid under investigation is put behind the Rochon polarizer and subsequently the transmitted spectrum is measured. First in a situation in which the Rochon polarizer and the polarizer under investigation are crossed, and then a situation in which they are parallel. The ratio of these two measurements then essentially yields the attenuation coefficient. Fig. (3.4) shows the results of such a measurement for a tandem wire grid and the squared results for a single wire grid. The two curves are very similar in appearance, although the curve for the tandem wire grid is systematically higher than the curve for the squared single wire grid. This small difference can be attributed to nonlinear behavior of the detection system. Note that below 3000 cm⁻¹ the curve for the tandem polarizer starts to rise. This is caused by the Rochon prism material, which becomes less and less birefringent as the wave number approaches the cutoff wave number of MgF₂ at 1500 cm⁻¹.

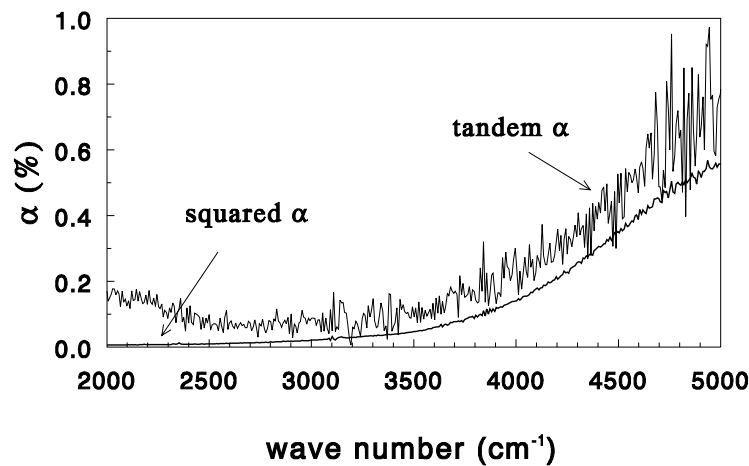


Figure 3.4: Squared attenuation coefficient α for a single wire grid polarizer and the attenuation coefficient for a tandem wire grid polarizer. The substrate material of the wire grid polarizers is KRS-5, and the wire spacing is $0.4 \mu\text{m}$.

A particular example of this nonlinear behavior in the detection system is the recently observed detector saturation that occurs at high intensities. This particularly affects the measurements with parallel transmission axes. The measured interferograms will be clipped at the center peak and therefore the resulting spectra will experience an offset. If a broader area around the center peak is clipped, an undulating effect might be observed in the spectrum besides the offset. An offset in the positive direction could well account for the observed systematic difference, while the undulation explains the rise of the tandem α below 3000 cm^{-1} .

Finally, it should be noted that a possible polarization sensitivity of the detector does not affect the measured values of the attenuation coefficient of the wire grid polarizers. Because the first Rochon polarizer may be considered perfect, only one polarization state is present in the optical system, i.e., a linear polarization state parallel to the transmission axis of the Rochon polarizer. The fixed position of the detector relative to the Rochon polarizer ensures that for both positions of the investigated wire grid (90° and 0° relative to the fixed polarizer), the intensity present in the linear polarization state is affected in the same way by a detector polarization. The ratio of the two intensities is thus independent of the detector polarization and is equal to the attenuation coefficient.

3.4 Source and detector imperfections

The Fourier transform spectrometer (Fig. (2.2)), serving as a light source for the ellipsometer, appears to introduce an elliptical polarization [17]. To describe this phenomenon the spectrometer is considered to be a virtual partial linear polarizer defined in Table (3.1) at position 2, placed behind a virtual unpolarized light

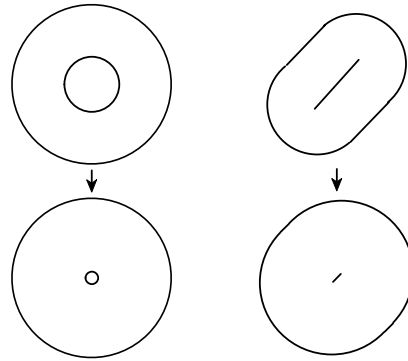


Figure 3.5: An intuitive interpretation of the polarization state of the light leaving the spectrometer. The outlines indicate the area swept through by the time varying E-field. The retardance is assumed to be 90° on the left side and 0° on the right side of the figure. The arrows indicate decreasing polarization degree.

source. This description is a simplification of reality. In fact the spectrometer should be represented by three parameters: an attenuating, a retarding, and an orientation parameter. As indicated in Fig. (3.5), for lower polarization degree the distinction between retardation and polarization degree becomes increasingly vague. Therefore a choice is made for this study to describe the source by two

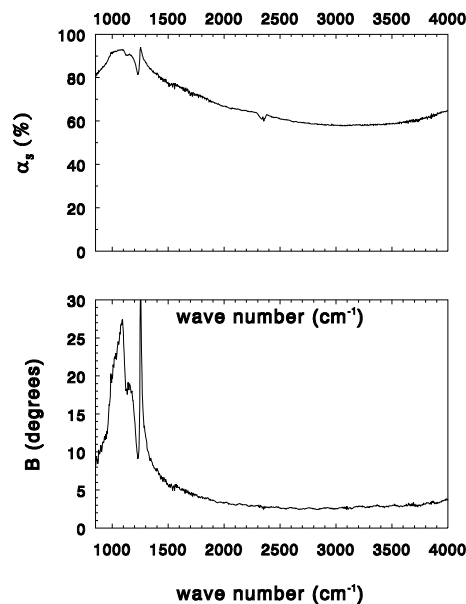


Figure 3.6: Attenuation coefficient α_s and azimuth B for the source. The graphs show artifacts from water (1200 cm^{-1} – 1900 cm^{-1}) and CO_2 (2350 cm^{-1}) in air. In regions of the spectra where the attenuation approaches 100% the accompanying azimuth becomes undefined.

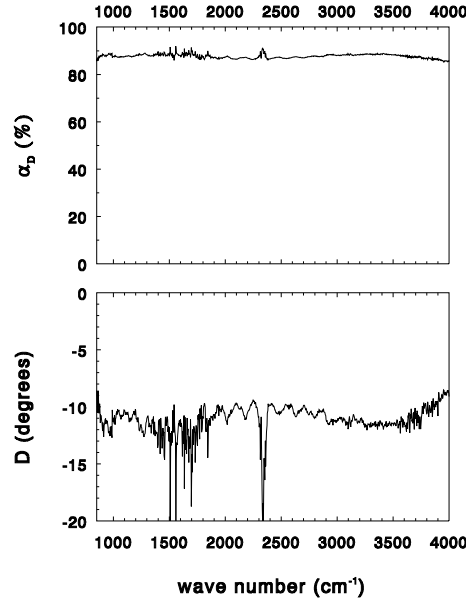


Figure 3.7: Attenuation coefficient α_D and azimuth D for the detector. The graphs show artifacts from water (1200 cm^{-1} – 1900 cm^{-1}) and CO_2 (2350 cm^{-1}) in air.

parameters, the attenuation coefficient α and the azimuth B .

To measure the source attenuation coefficient and spatial orientation, two sets of double wire grids with transmission axes parallel are placed behind the spectrometer. The second set of wire grids serves to eliminate a possible sensitivity of the detector to polarization. The angle between the spectrometer and the two sets of wire grids is denoted B . Now the first set of wire grids is rotated through 360° in steps of 3° and again for each position a spectrum is measured. From cross sections through the spectra Fourier coefficients may be calculated as a function of wavelength. Evaluation of the product of the Mueller matrices representing the components in the setup yields relations between the Fourier coefficients and the desired quantities α_S and B . With these relations the spectra of the source attenuation coefficient α_S and the source spatial orientation B may be calculated from the Fourier coefficients. Results are exhibited in Fig. (3.6).

Another imperfect component in the setup is the MCT detector. It appears to be sensitive to the state of polarization of the incident light. This is described by placing a virtual partial polarizer in front of a virtual polarization insensitive detector. Again the attenuation coefficient and spatial orientation are measured by placing two sets of double wire grids with transmission axes parallel. The angle between the virtual polarizer of the detector and the two sets of wire grids is denoted D . Now the second set of wire grids is rotated through 360° in steps of 3° and in a fashion similar to the calculation of the source characteristics the detector attenuation coefficient α_D and the detector spatial orientation D may be obtained as a function of wavelength. Results are exhibited in Fig. (3.7).

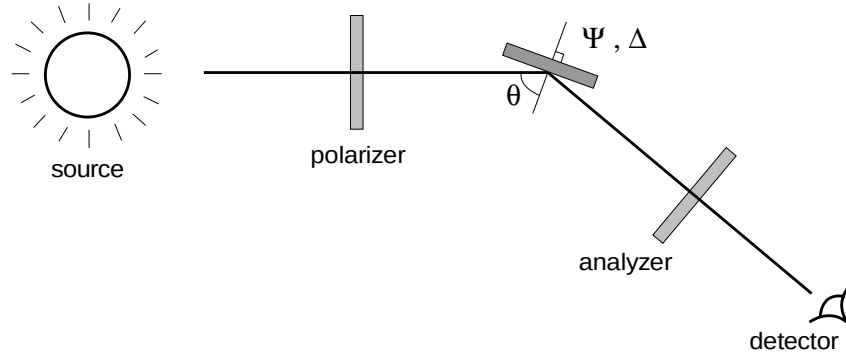


Figure 3.8: Rotating analyzer setup containing only perfect components. The angles relative to the plane of incidence are denoted by P for the polarizer and A for the analyzer.

3.5 Calculating the rotating analyzer ellipsometer behavior

Knowledge of the intensity arriving at the detector as a function of all imperfections and component orientations is essential for the correct determination of the ellipsometric angles Ψ and Δ . To calculate the intensity, the Mueller matrices, outlined in the previous sections, are multiplied and applied to a Stokes vector which describes the light source. This section deals with two rotating analyzer ellipsometers: a setup containing only perfect components and a setup consisting of imperfect components. For both setups a relation between the measurable Fourier coefficients and the desired ellipsometric angles Ψ and Δ is derived.

3.5.1 Perfect rotating analyzer setup

Setting out from the unpolarized light source, represented by the well known Stokes vector $\underline{\mathbf{S}}_0 = (1, 0, 0, 0)$, the light passes the following train of optical components depicted in Fig. (3.8):

- A perfect polarizer, represented by the Mueller matrix in Table (3.1) at position 1, and azimuth P

$$\underline{\mathbf{R}}(-P) \cdot \underline{\mathbf{M}}_P \cdot \underline{\mathbf{R}}(P). \quad (3.6)$$

The Mueller matrix for the perfect polarizer is pre- and post-multiplied by a rotation matrix $\underline{\mathbf{R}}$, in order to bring the component into a reference frame rotated over an angle P relative to the plane of incidence.

- The sample under investigation, characterized by Ψ and Δ and represented by the Mueller matrix from Table (3.1) at position 5. The sample defines the plane of incidence, and is therefore not multiplied by any rotation matrices.

- A perfect analyzer, represented by the same Mueller matrix as the polarizer. Now this matrix is pre- and post-multiplied by rotation matrices, bringing the component to an angle A relative to the plane of incidence.

$$\underline{\underline{\mathbf{R}}}(-A) \cdot \underline{\underline{\mathbf{M}}}_A \cdot \underline{\underline{\mathbf{R}}}(A). \quad (3.7)$$

Multiplication of the initial unpolarized Stokes vector with this series of Mueller matrices yields for the resulting Stokes vector

$$\underline{\underline{\mathbf{S}}}_u = \underline{\underline{\mathbf{M}}}_A \cdot \underline{\underline{\mathbf{R}}}(A) \cdot \underline{\underline{\mathbf{M}}}_{\Psi\Delta} \cdot \underline{\underline{\mathbf{R}}}(-P) \cdot \underline{\underline{\mathbf{M}}}_P \cdot \underline{\underline{\mathbf{S}}}_0 \quad (3.8)$$

Observe the absence of the rotation matrices $\underline{\underline{\mathbf{R}}}(P)$ and $\underline{\underline{\mathbf{R}}}(-A)$ at the extreme right and left sides of Eq. (3.8). As the incident unpolarized Stokes vector is invariant under rotation of the frame of reference, the first rotation matrix may be left out. Similarly, the perfect detector, i.e., a detector measuring only the intensity present in a polarization state indifferent to the polarization state, permits removal of the preceding rotation matrix. A rotation matrix only changes the orientation of the polarization state, not the intensity.

The first component of the resulting Stokes vector in Eq. (3.8) represents the intensity at the detector. The intensity, I_D , can be written as

$$I_D = (1 - \cos 2\Psi \cos 2P) \left[1 + \frac{\cos 2P - \cos 2\Psi}{1 - \cos 2\Psi \cos 2P} \cos 2A + \frac{\cos \Delta \sin 2\Psi \sin 2P}{1 - \cos 2\Psi \cos 2P} \sin 2A \right]. \quad (3.9)$$

It is obvious that the intensity is a sinusoidal function of the analyzer angle A and may be written in the form

$$I_D = g (1 + a \cos 2A + b \sin 2A), \quad (3.10)$$

where, from Eq. (3.9), the Fourier coefficients are easily seen to be

$$\begin{aligned} g &= 1 - \cos 2\Psi \cos 2P, \\ a &= \frac{\cos 2P - \cos 2\Psi}{1 - \cos 2\Psi \cos 2P}, \\ b &= \frac{\cos \Delta \sin 2\Psi \sin 2P}{1 - \cos 2\Psi \cos 2P}. \end{aligned} \quad (3.11)$$

The ellipsometric angles Ψ and Δ can be expressed explicitly in these Fourier coefficients

$$\begin{aligned} \cos 2\Psi &= \frac{\cos 2P - a}{1 - a \cos 2P}, \\ \cos \Delta &= \frac{b}{\sqrt{1 - a^2}} \cdot \frac{\sin 2P}{|\sin 2P|}. \end{aligned} \quad (3.12)$$

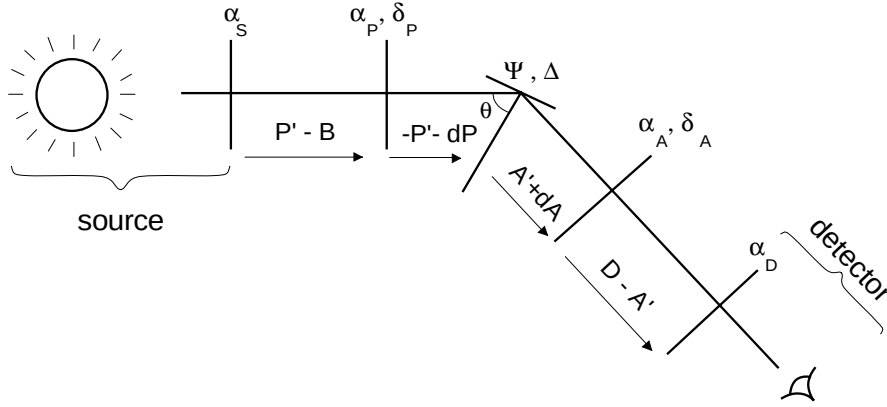


Figure 3.9: Setup containing a partially polarized light source, partially polarizing and retarding polarizers, and a detector that is sensitive to the state of polarization of incident light. Angles between components are indicated.

In experiments the polarizer is usually put at an angle of 45° relative to the plane of incidence. Substituting this value for P in Eq. (3.12) yields the familiar rotating analyzer expressions for Ψ and Δ in terms of the measured Fourier coefficients

$$\begin{aligned} \cos 2\Psi &= -a, \\ \cos \Delta &= \frac{b}{\sqrt{1 - a^2}}. \end{aligned} \quad (3.13)$$

3.5.2 Imperfect setup

Anticipating the calibration procedure in the following section, two error angles are introduced

$$\begin{aligned} A &\equiv A' + dA & \text{and} \\ P &\equiv P' + dP. \end{aligned} \quad (3.14)$$

Here the unprimed symbols represent the real position of the analyzer and the polarizer relative to the plane of incidence, the primed symbols are the positions as they are read from the rotation supports on which the components are mounted, and dA and dP are the differences between the two positions.

In common with the perfect setup the calculation for the imperfect setup starts with the unpolarized light source. The emitted light passes the following train of optical components depicted in Fig. (3.9):

- A partial polarizer from Table (3.1) at position 2 with attenuation coefficient α_S and azimuth B given in Fig. (3.6). This component represents the

partially polarizing properties of the source as described in the previous section. Note that the angle B is defined relative to the rotation support of the following component. Therefore the error angle dP must be added to the source azimuth B . Thus the total Mueller matrix for this component can be written as the product of two rotation matrices and a partial polarizer matrix

$$\underline{\underline{\mathbf{R}}}(-B - dP) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_S} \cdot \underline{\underline{\mathbf{R}}}(B + dP). \quad (3.15)$$

- The polarizer in Eq. (3.3) with attenuation coefficient α_P (Fig. (3.2a)), retardance δ_P (Fig. (3.2b)), and azimuth $P' + dP$.

$$\underline{\underline{\mathbf{R}}}(-P' - dP) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_P \delta_P} \cdot \underline{\underline{\mathbf{R}}}(P' + dP). \quad (3.16)$$

- The sample characterized by the ellipsometric angles Ψ and Δ represented by the Mueller matrix from Table (3.1), position 5.
- The analyzer in Eq. (3.3) with attenuation coefficient α_A (Fig. (3.2a)), retardance δ_A (Fig. (3.2b)), and azimuth $A' + dA$.

$$\underline{\underline{\mathbf{R}}}(-A' - dA) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_A \delta_A} \cdot \underline{\underline{\mathbf{R}}}(A' + dA). \quad (3.17)$$

- A partial polarizer representing the partially polarizing properties of the detector. Attenuation coefficient α_D and azimuth D are given in Fig. (3.7). Note that the angle D is defined relative to the preceding analyzer. Therefore, similar to the source, an error angle dA has to be added to the azimuth angle D .

$$\underline{\underline{\mathbf{R}}}(-D - dA) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_D} \cdot \underline{\underline{\mathbf{R}}}(D + dA). \quad (3.18)$$

Thus the resulting state of polarization can be expressed as the Stokes vector $\underline{\mathbf{S}}_u$

$$\begin{aligned} \underline{\mathbf{S}}_u = & \underline{\underline{\mathbf{M}}}_{\alpha_D} \cdot \underline{\underline{\mathbf{R}}}(D - A') \cdot \underline{\underline{\mathbf{M}}}_{\alpha_A \delta_A} \cdot \underline{\underline{\mathbf{R}}}(A' + dA) \cdot \underline{\underline{\mathbf{M}}}_{\Psi \Delta} \cdot \\ & \underline{\underline{\mathbf{R}}}(-P' - dP) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_P \delta_P} \cdot \underline{\underline{\mathbf{R}}}(P' - B) \cdot \underline{\underline{\mathbf{M}}}_{\alpha_S} \cdot \underline{\mathbf{S}}_0. \end{aligned} \quad (3.19)$$

Note again the absence of the rotation matrices $\underline{\underline{\mathbf{R}}}(B + dP)$ and $\underline{\underline{\mathbf{R}}}(-D - dA)$ which are left out on the same grounds as in the case of the perfect setup. Observe, furthermore, that multiplication of rotation matrices results in the addition of the rotation angles.

The intensity on the detector, the first component of the resulting Stokes vector in Eq. (3.19), may be expressed as a sinusoidal function

$$I_D = g(1 + a \cos 2A' + b \sin 2A'). \quad (3.20)$$

From Fourier analysis of the intensity as a function of the analyzer angle A' the Fourier coefficients a and b can be calculated

$$\begin{aligned}
g &= \frac{1}{\pi} \int_0^{\pi} I_D(A') \, dA', \\
a &= \frac{2}{g} \int_0^{\pi} I_D(A') \cos 2A' \, dA', \\
b &= \frac{2}{g} \int_0^{\pi} I_D(A') \sin 2A' \, dA',
\end{aligned} \tag{3.21}$$

and, since all calculations are performed analytically, a relation between (a, b) and (Ψ, Δ) plus all system parameters may be established. Once a, b and all system parameters are determined, inversion of the relation yields Ψ and Δ .

3.6 Calibration of polarizer and analyzer angles in a spectroscopic ellipsometer

Reliable operation of any ellipsometer stands or falls with the accurate determination of the positions of all of its components relative to the plane of incidence. Incorrect positioning of e.g. the polarizer is directly translated to erroneous values of Ψ and Δ [18]. For this reason the calibration procedure is one of the most important steps in preparation for ellipsometric experiments. For single wavelength ellipsometers several complementary calibration procedures are available [19–22]. Best known procedures are the residue and phase calibration procedures. The objective of this section is to establish a calibration procedure for spectroscopic rotating analyzer ellipsometers. Such a procedure has two mainstays. The first is the residue calibration method for a single wavelength rotating analyzer ellipsometer. The second is the fact that the calibration angles are geometric objects. This implies that the angles do not depend on wavelength, which makes wavelength independence a valuable criterion in finding a spectroscopic calibration procedure. A further demand made on the calibration procedure is that the results of the calibration procedure do not depend on the sample parameters and all other system parameters.

3.6.1 Determination of the polarizer angle

As a starting-point the residue is defined, slightly differently from usual [21, 22], as

$$R \equiv a^2 + b^2, \tag{3.22}$$

where a and b are the Fourier coefficients from Eq. (3.21). This somewhat different definition permits analytical calculations, while it does not greatly affect the behavior of the residue. The residue is defined such that it comprises both measured quantities a and b , so that any variation in either of these two Fourier

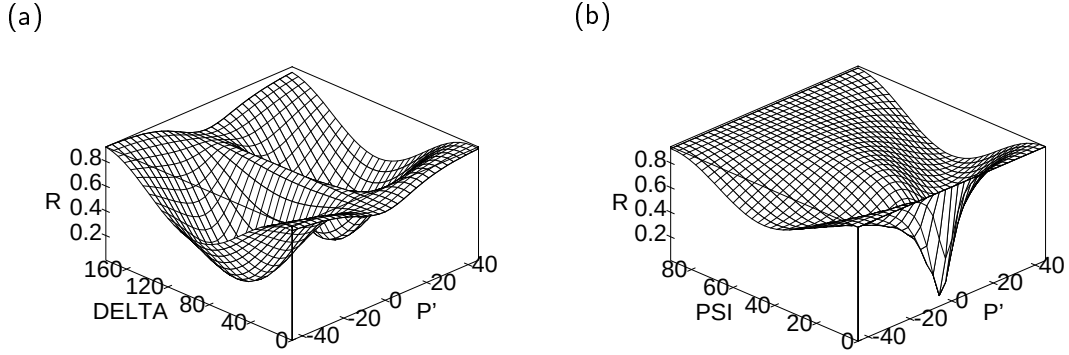


Figure 3.10: (a) Residue as a function of Δ and P' for $\Psi = 22.5^\circ$ and $dP = 0$. (b) Residue as a function of Ψ and P' for $\Delta = 150^\circ$ and $dP = 0$.

coefficients will show up in the residue. In traditional single wavelength ellipsometry the residue is minimized as a function of polarizer angle in order to find the polarizer angle relative to the plane of incidence.

The residue is plotted, for realistic system parameters, versus Δ ($\Psi = 22.5^\circ$) and P' in Fig. (3.10a), and versus Ψ ($\Delta = 150^\circ$) and P' in Fig. (3.10b). In both cases $dP = 0$. For other fixed values of Ψ and Δ , the pictures change only slightly. From Fig. (3.10a) it is clear that in the plane of incidence ($P' = 0$), the residue is independent of Δ , whereas from Fig. (3.10b) can be seen that for $\Psi > 30^\circ$ the residue shows only a small dependence on Ψ . Further calculations show that the polarizer position for the maximum of the residue in Fig. (3.10a) changes significantly only when Δ assumes either very small or very high values. These observations lead to the following criterium: taking a sample with Ψ more or less constant at a value greater than 30° for all wavelengths, and Δ varying as a function of wavelength over a large range, the residue will exhibit a strong variation as a function of wavelength, for all angles P' , except when P' is in the plane of incidence. So, by minimizing the variation in the residue as a function of wavelength, the plane of incidence, i.e., $P = 0$ and thus dP , may be determined. A sample that satisfies the above-mentioned requirements is a gold substrate with a thick transparent layer on top of it. In that case Ψ will be between 40° and 45° and Δ will vary over a large range.

3.6.2 Determination of the analyzer angle

In this case we set out from the Fourier coefficient a in Eq. (3.21), and make plots of a , again for realistic system parameters, versus Δ ($\Psi = 45^\circ$) and dA in Fig. (3.11a), and versus Ψ ($\Delta = 150^\circ$) and dA in Fig. (3.11b). Values $\Psi = 45^\circ$ and $\Delta = 150^\circ$ are chosen because they are typical values for calibration measurements on samples as discussed in the previous section. From Fig. (3.11a) and Fig. (3.11b) it can be seen that a has a maximum independent of Δ , and is only slightly dependent on Ψ . The location of the maximum, ϵ , may now be determined by varying dA . The value of ϵ is small and depends only on the component

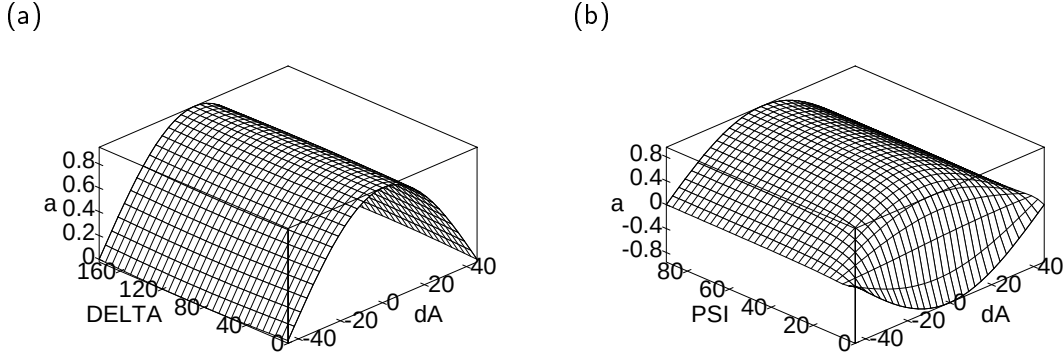


Figure 3.11: (a) Fourier coefficient a as a function of Δ and dA for $\Psi = 45^\circ$ and $P = 0$. (b) Fourier coefficient a as a function of Ψ and dA for $\Delta = 150^\circ$ and $P = 0$.

imperfections.

In practice it is easier to determine the analyzer position from the knowledge that when the polarizer is in the plane of incidence, only a linear polarization state parallel to the plane of incidence propagates through the ellipsometer. In this case Fourier coefficient a should be 1 and b should be 0. From these values and the measured values of a and b the analyzer error angle may be calculated as follows

$$dA = -\frac{1}{2} \arctan \frac{b}{a}. \quad (3.23)$$

This relation is derived from the transformation of Fourier coefficients caused by an error angle of the rotating component. More on this topic can be found in Section (5.4).

3.7 Conclusion

In this chapter the imperfections present in a rotating analyzer ellipsometer have been described, using the Mueller matrix representation, and measured. Thus the source, detector, and especially the wire grid polarizers have been characterized. Based on the results for a single wire grid polarizer, a better tandem wire grid has been developed. This latter polarizer performs quadratically better, whereas it does not greatly reduce the transmitted intensity. The reduction in intensity caused by reflection losses could even be improved upon in future designs of grid polarizers by mounting the grid on both sides of the substrate.

Making use of the Mueller description for all imperfect components, the behavior of the ellipsometer could be calculated. A calibration procedure for the ellipsometer was derived from these calculations.

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Chapter 4

Measurement of the complex refractive index of liquids in the infrared

Abstract

With spectroscopic ellipsometry one can measure the real and imaginary parts of the refractive index of a medium simultaneously. To determine this index in the infrared for a number of technologically important liquids, use was made of attenuated total internal reflection at the glass-liquid interface of a specially designed prism. This ATR approach warrants minimal signal loss and is, for strongly absorbing liquids, the only way to measure the complex refractive index. A surprising phenomenon, observed when BK-7 prism glass was used, is the scattering in the vicinity of the absorption wavelengths of the glass. A simple model that can be used to describe the relations among absorption, scattering, and depolarization was successfully used to correct the measurements. Refractive indices for demineralized water, Freon 113, heptane, benzene, gas oil, and crude oil in the wave number range from 5000 to 10000 cm^{-1} (1–2 μm) are presented.

4.1 Introduction

Most methods that determine the refractive index of a liquid can be used to measure only either the real or the imaginary part. One can usually determine the imaginary part using an absorption measurement [1, 2] through a sample of known thickness. For liquids with a high absorption coefficient this means that the samples must be very thin to measure any intensity at all. However, the required thickness and parallelism of front and backside of a sample cause multiple reflections to occur and therefore limit the accuracy. The problem of absorption becomes even more acute when one determines the real part of the refractive index using a refractometer [3], an interferometric technique [4, 5], or a beam deviation technique [6, 7]. In this case the light travels a relatively long distance

in the material and almost the entire intensity may be absorbed. Furthermore, in the case of absorption the angle of refraction depends on both the real and imaginary parts of the refractive index, so it is impossible to distinguish between them.

One method that is capable of measuring the real and imaginary parts of the refractive index of a liquid simultaneously is ellipsometry. Havstad *et al.* [8] thus measured the refractive index of liquid uranium under ultra high vacuum conditions for wavelengths in the range from 400 nm to 10 μm . Brückner *et al.* [9] did the same for liquid gold and tin at 10.6 μm . The problem with these measurements is that the free liquid–vacuum interface is sensitive to vibrations from the environment, and, more importantly, for materials other than metals most of the incident intensity is lost as it is transmitted into the liquid.

To solve this problem an ATR-like method [10] is presented, in which the liquid is locked behind a glass prism in a sample holder and ellipsometry is performed on the stable glass-liquid interface. Besides the gain in stability and ease of handling the major advantage is the possibility to measure under the condition of total internal reflection. If the real part of the refractive index of the prism is higher than that of the liquid and the angle of incidence is sufficiently high, the full intensity is reflected toward the detector.

4.2 Theoretical background

4.2.1 Ellipsometry

As calculated in Section (3.5) for a perfect rotating analyzer ellipsometer setup, the relations between the ellipsometric angles Ψ and Δ on the one hand and the measured Fourier coefficients a and b on the other hand are given by Eq. (3.13)

$$\begin{aligned}\cos 2\Psi &= -a, \\ \cos \Delta &= \frac{b}{\sqrt{1 - a^2}}.\end{aligned}\tag{4.1}$$

Note that these equations are valid only if the polarizers are perfectly linearly polarizing, the detector is equally sensitive to all polarization states, the transmission axis of the polarizer makes an angle of 45° with the plane of incidence and the starting position of the transmission axis of the analyzer is in the plane of incidence. When polarizer, source, and detector imperfections are taken into account, as is done for the imperfect setup in Section (3.5), the above expressions become implicit and have to be solved numerically for Ψ and Δ [11].

4.2.2 Relation between the refractive index and (Ψ, Δ)

For reflection at an interface between two semi-infinite media Fresnel defined two reflection coefficients presented in Eqs. (2.1). As stated in Section (2.2), the ratio

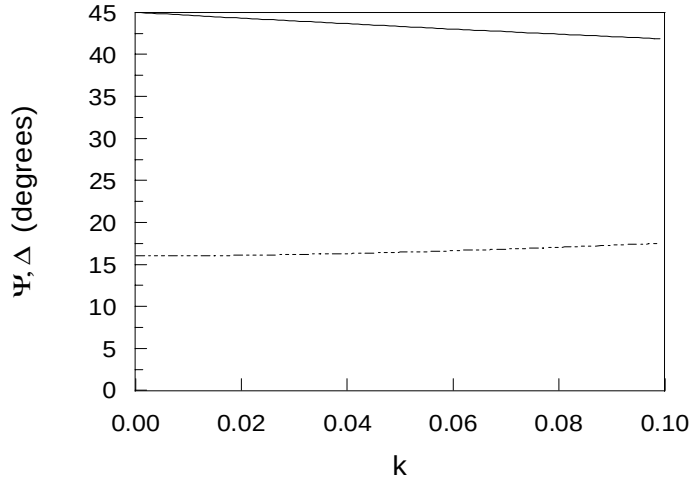


Figure 4.1: Ψ (solid line) and Δ (dashed line) versus the imaginary part k of the refractive index of the liquid n_1 . The refractive index n_0 of the prism is assumed to be real and has a value of 1.5. The real part of the refractive index of the liquid is 1.3. The angle of incidence is 70° and the critical angle is 60° . The value of Δ is insensitive to small values of the imaginary part of the refractive index, whereas Ψ exhibits a linear dependence for small values of k .

of these two coefficients defines in Eq. (2.2) a complex quantity ρ that may be interpreted as the change of polarization on reflection. The ellipsometric angles Ψ and Δ are derived from this complex quantity ρ by writing it in polar notation and represent the relative changes in amplitude and phase upon reflection, as was already established in Section (2.2). From the reflection coefficients, the change in polarization state ρ , and Snell's law

$$n_0 \sin \theta_0 = n_1 \sin \theta_1, \quad (4.2)$$

there follows an explicit expression for the refractive index of the second medium [12]

$$n_1 = n_0 \sin \theta_0 \sqrt{1 + \left(\frac{1 - \rho}{1 + \rho} \right)^2 \tan^2 \theta_0}. \quad (4.3)$$

After Ψ and Δ are determined for a specific angle of incidence θ_0 and when the refractive index of the incident medium is known, Eq. (4.3) permits the calculation of the refractive index of the other medium.

To gain insight into the relation in Eq. (4.3) between the refractive indices and (Ψ, Δ) , consider two extreme cases in which the real part of the refractive index of the incident medium is larger than that of the second medium. First of all the situation of total internal reflection is investigated. When both media are non-absorbing and the incident angle is larger than the critical angle, the light is totally internally reflected. If this situation occurs, $|r_p| = |r_s| = 1$ and obviously

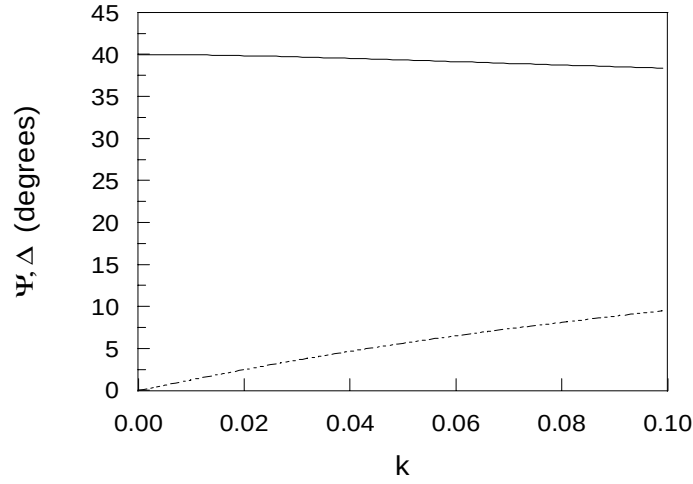


Figure 4.2: Ψ (solid line) and Δ (dashed line) versus the imaginary part k of the refractive index of the liquid n_1 . The refractive index n_0 of the prism is assumed to be real and has a value of 1.5. The real part of the refractive index of the liquid is 1.45. The angle of incidence is 70° and the critical angle is 75° . The value of Ψ is insensitive to small values of the imaginary part of the refractive index, whereas Δ exhibits a linear dependence for small values of k .

$\Psi = 45^\circ$, so that $\rho = \exp j\Delta$. This simplifies Eq. (4.3) to a point at which Δ can be expressed in terms of the incident angle and the refractive indices of the prism and the liquid [13]

$$\tan \left| \frac{1}{2} \Delta \right| = \frac{\cos \theta_0 \sqrt{\sin^2 \theta_0 - \left(\frac{n_1}{n_0} \right)^2}}{\sin^2 \theta_0}. \quad (4.4)$$

If the second medium is slightly absorbing, it can be shown that no total internal reflection can occur. For large angles of incidence, though, there is attenuated total reflection. It is clear from Eq. (4.4) that also in this case a change in Δ relates closely to a change in the real part of the refractive index. Similarly, a deviation of Ψ from 45° relates to the imaginary part of the refractive index. This relation between the imaginary part of the refractive index and Ψ is illustrated in Fig. (4.1).

The second extreme case arises when the angle of incidence is smaller than the critical angle and again both media are non-absorbing. If the angle of incidence is still higher than the Brewster angle, $\Delta = 0^\circ$; otherwise $\Delta = 180^\circ$. In both cases the ratio ρ of the reflection coefficients will be real and equal to $\pm \tan \Psi$, leading to

$$\tan \Psi = \pm \frac{\sin^2 \theta_0 - \cos \theta_0 \sqrt{\left(\frac{n_1}{n_0} \right)^2 - \sin^2 \theta_0}}{\sin^2 \theta_0 + \cos \theta_0 \sqrt{\left(\frac{n_1}{n_0} \right)^2 - \sin^2 \theta_0}}, \quad (4.5)$$

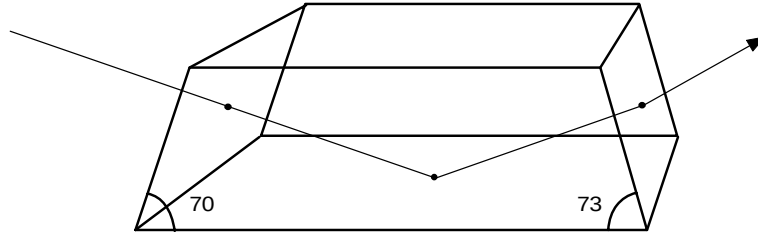


Figure 4.3: BK-7 prism, the light is incident through the left side, reflects at the fluid-glass interface at the bottom and leaves the prism through the right side. A small difference in angle between the left and the right sides prevents multiple reflections in the prism.

with

$$\begin{cases} + : \Delta = 0^\circ & \arctan(n_1/n_0) \leq \theta_0 < \arcsin(n_1/n_0), \\ - : \Delta = 180^\circ & \theta_0 < \arctan(n_1/n_0). \end{cases}$$

If the second medium is slightly absorbing, it can be shown that the value of Δ deviates slightly from either 0° or 180° . It is clear from Eq. (4.5) that, as long as this deviation is small, a change in Ψ will relate to a change in the real part of the refractive index. Similarly the deviation of Δ from 0° (or 180°) will relate to the imaginary part of the refractive index. This relation between the imaginary part of the refractive index and Δ is illustrated in Fig. (4.2)

From these two extreme cases one should note the contrast in the role of the ellipsometric quantities Ψ and Δ . In the case of attenuated total reflection the real part relates to Δ and the imaginary part relates to Ψ , whereas in the other case — two real refractive indices and an angle of incidence below the critical angle — the real part relates to Ψ and the imaginary part to Δ .

4.2.3 Prism

To avoid vibrations from the environment disturbing the liquid surface a prism with a known refractive index is used to cover the liquid. This prism depicted in Fig. (4.3) at the same time provides several other advantages. First, it permits the glass-liquid interface to be placed at any desired orientation. It also facilitates handling, prevents pollution of the liquid, and keeps the liquid from evaporating, thus not requiring extensive safety measures with the more dangerous liquids such as benzene. But the most important advantage is that the real part of the refractive index of the prism is higher than that of the liquid so that for large angles of incidence a situation of attenuated total internal reflection arises. This ensures a high intensity at the detector.

The incident light beam enters the prism perpendicularly. Both the parallel and perpendicular component experience the same amplitude change so that their ratio remains unchanged. Neither of the components shifts in phase, so also the relative phase is unchanged.

After entering the prism the beam reflects at the glass-liquid interface at the backside of the prism. This is described by Eq. (4.3). To prevent multiple reflections in the prism, it was designed such that the light that leaves the prism strikes the glass-air interface at an angle that is slightly off-perpendicular. This has the side effect that it also introduces a relative amplitude change that is given by the Fresnel transmission coefficients

$$\begin{aligned} t_p &= \frac{2n_0}{n_1 \cos \theta_0 + n_0 \cos \theta_1}, \\ t_s &= \frac{2n_0}{n_0 \cos \theta_0 + n_1 \cos \theta_1}. \end{aligned} \quad (4.6)$$

Note that n_0 and n_1 now represent the refractive indices of the prism glass and air, respectively. Together with Snell's law (Eq. (4.2)), for the relative amplitude change this gives

$$\begin{aligned} \tan \Psi_t &\equiv \left| \frac{t_p}{t_s} \right| \\ &= \frac{n_0 n_1 \cos \theta_0 + n_1 \sqrt{n_1^2 - n_0^2 \sin^2 \theta_0}}{n_1^2 \cos \theta_0 + n_0 \sqrt{n_1^2 - n_0^2 \sin^2 \theta_0}}. \end{aligned} \quad (4.7)$$

The phase remains unchanged.

The total effective relative amplitude change that the light has experienced in the prism is now given by

$$\tan \Psi_{\text{eff}} = \tan \Psi \tan \Psi_t. \quad (4.8)$$

The angle Ψ_{eff} is measured and Ψ_t can be calculated by Eq. (4.7). Thus Ψ can be calculated from the measured Ψ_{eff} and the calculated Ψ_t by Eq. (4.8). This Ψ together with Δ — equal to the measured Δ_{eff} — can be inserted into Eq. (2.2) to calculate the complex quantity ρ . This, in combination with the refractive index of the prism, the angle of incidence, and Eq. (4.3), permits calculation of the refractive index of the liquid.

4.3 Experimental setup

4.3.1 Rotating analyzer ellipsometer

The light source used in this experiment (Fig. 4.4) is a cascaded arc, which emits from the ultraviolet to the infrared parts of the spectrum [14]. By means of a mirror the light coming from the arc is coupled into a Bruker IFS66 Fourier transform infrared (FTIR) spectrometer. The spectrometer contains a CaF_2 beam splitter that determines the lower limit of the wave number range to 1650 cm^{-1} . The light that emerges from the FTIR spectrometer then strikes a MgF_2 Rochon polarizer, passes through a diaphragm that blocks the extraordinary beam that also

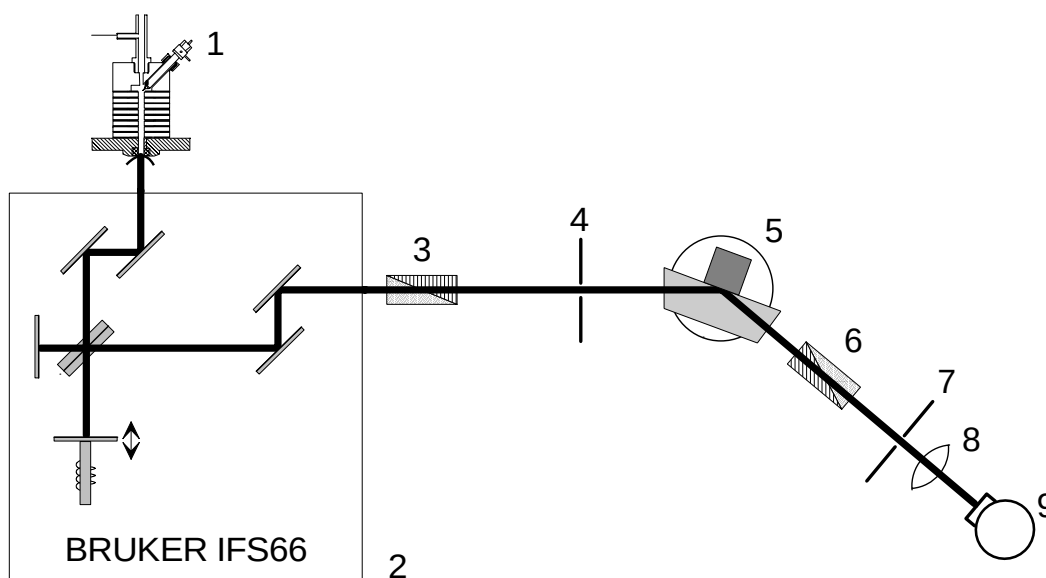


Figure 4.4: Spectroscopic ellipsometer setup with 1, cascaded arc light source; 2, Bruker IFS66 spectrometer; 3, Rochon prism polarizer; 4, diaphragm; 5, sample holder and prism; 6, Rochon prism polarizer; 7, diaphragm; 8, imaging lens; 9, HgCdTe detector.

emerges from the Rochon polarizer, and enters the prism where it reflects at the glass-liquid interface. After the light leaves the prism it falls on the analyzer, which is identical to the polarizer, passes another diaphragm and a lens, and finally hits a HgCdTe detector. This detector fixes the upper limit of the wave number range at approximately 10000 cm^{-1} .

The polarizer and analyzer are mounted on stepper motor controlled rotation supports. The polarizer is held fixed with the transmission axis at an angle of 45° relative to the plane of incidence. The analyzer is rotated in steps. For each step a spectrum is measured. From a cross-section through these spectra at a fixed wave number the Fourier coefficients may be determined and, hence, Ψ and Δ and the refractive index.

The sample holder is placed such that the incident light strikes the prism perpendicularly. In that case the angle of incidence on the glass-liquid interface is 70° . Prior to measurement the position of both polarizer and analyzer with respect to the plane of incidence needs to be determined. This is done by the method described in Section (3.6).

4.3.2 Liquid sample holder and prism

The liquids are contained in an aluminum sample holder with a volume of approximately 20 cm^3 . The sample holder is closed by the prism and sealed by an

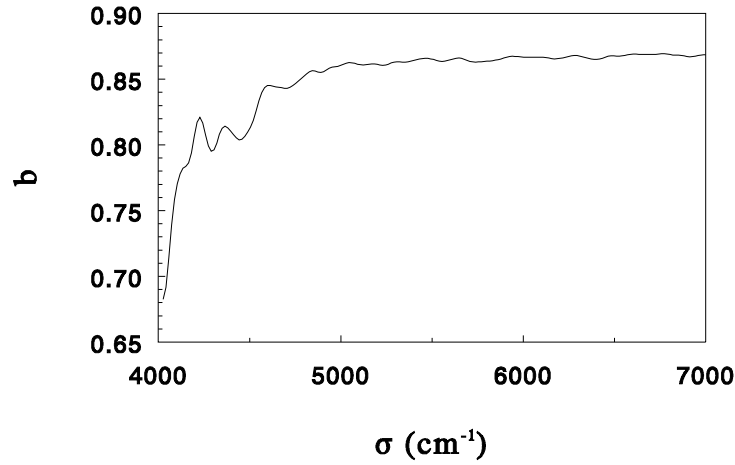


Figure 4.5: Fourier coefficient b versus wave number calculated from a measurement with air behind the prism. Note the sudden drop in value in the vicinity of 5000 cm^{-1} . The average value of b in an undisturbed range of the spectrum is 0.8662. This is 0.01 below the calculated value for a BK-7–air interface. However, this measurement was performed with the ellipsometer not yet fully calibrated, i.e., the angles of polarizer and analyzer relative to the plane of incidence were not yet exactly determined.

O-ring. The prism material is BK-7 glass manufactured and certified by Schott Glaswerke. The refractive index of BK-7, close to 1.5 and expressed in a Sellmeier dispersion formula [15], is accurate to the fifth decimal. The BK-7 prism further limits the wave number range from approximately 5000 cm^{-1} upward.

Liquids that have been investigated are water, Freon, heptane, benzene, gas oil and crude oil.

4.4 Results

4.4.1 Scattering

For wave numbers that approach the cutoff wavelength of the prism BK-7 glass (5000 cm^{-1}), spectra of the refractive index show a sudden drop in value. This decrease is specifically observed for the real part of the refractive index and can be traced back to a similar decrease in the Fourier coefficient b (Fig. 4.5). As this phenomenon is independent of the liquid under observation and coincides with the cutoff wavelength of BK-7 glass, it must be attributed to absorption in the BK-7 glass. Further proof of this assumption is obtained from an infrared transmission spectrum of BK-7 glass shown in Fig. (4.6), which reveals a similar structure as observed in the real parts of the refractive index spectra. This observation also rules out a possible surface phenomenon.

Absorption on its own is not sufficient to explain the decrease in refractive

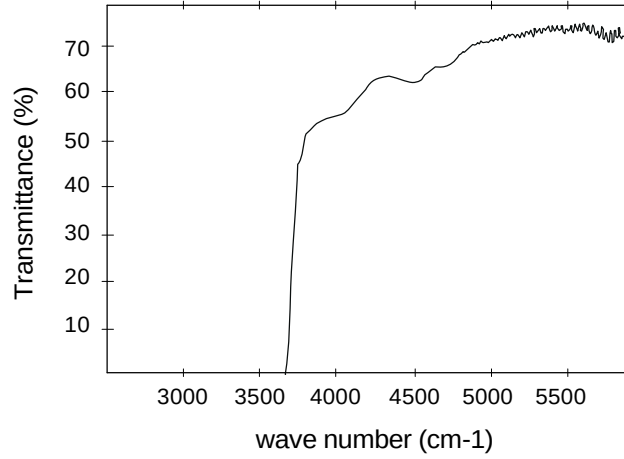


Figure 4.6: The transmission of a piece of BK-7 glass versus wave number. A structure similar to the one observed in the refractive index spectra can be seen. The noise at the right side of the spectrum is caused by a decreasing intensity. The beam splitter (KBr) used in the spectrometer becomes opaque for wave numbers in excess of approximately 5000 cm^{-1} .

index values, but absorption is usually accompanied by scattering. Scattering lowers the polarization degree of the light that reflects at the sample, and as such disturbs the ellipsometric measurement [16]. A decreased polarization degree Π leads to Fourier coefficients a^* and b^* related to the undisturbed coefficients a and b in Eqs. (4.1) through

$$\begin{aligned} a^* &= \Pi a = -\Pi \cos 2\Psi, \\ b^* &= \Pi^2 b = \Pi^2 \cos \Delta \sin 2\Psi. \end{aligned} \quad (4.9)$$

These relations are obtained from complete evaluation of a perfect rotating analyzer ellipsometer in common with Section (3.5), into which depolarizing components in front of and behind the reflecting sample are introduced. The depolarizing components are represented by matrices from Table (3.1) position 6, where the depolarization coefficient is equal to Π . It is obvious that the b coefficient is affected more strongly than the a coefficient, especially when $a \approx 0$ and $b \approx 1$ as is the case with attenuated total reflection.

To describe the scattering a simple model, derived from a description of scattering by small spherical particles by Van de Hulst [17], is used. For small particles with a small absorption coefficient one may assume that the coefficient for scattering Q has a linear relation with the wave number σ and the imaginary part of the refractive index k

$$Q = 2\pi r \xi \sigma k. \quad (4.10)$$

Here r is the radius of the scattering particles and ξ is a constant factor. As the absorption and subsequent scattering increase, the polarization degree decreases.

When the polarization degree drops to a value of 0.9 the absorption has become so high that no more light reaches the detector. This means that only values of Π close to 1, that is small values of the scattering coefficient, are of interest. For small values of the scattering coefficient the relation with the polarization degree can be approximated by any power series that satisfies the following two boundary conditions: the polarization degree must decrease monotonically with increasing scattering coefficient, and for zero scattering coefficient the polarization degree must be unaffected. As a first estimate an exponential function is chosen to reflect the relation

$$\Pi = e^{-Q}. \quad (4.11)$$

For zero scattering coefficient the polarization degree is indeed unaffected whereas for large scatter coefficient the light depolarizes completely ($\Pi=0$).

To describe the cutoff of the BK-7 prism glass at 5000 cm^{-1} , it is necessary to realize that the absorption is caused by vibration of several chemical bonds. Absorption of this kind can be described by a harmonic oscillator model resulting in a Lorentz profile. The Lorentz profile is characterized by three parameters. These parameters are the absorbing wave number, the width and the strength of the absorption. In this case, though, more than one absorption plays a role, so the relation between the imaginary part of the refractive index of BK-7, k , and the wave number σ , is the sum of several Lorentz profiles:

$$k(\sigma) = \sum_{i=1}^6 \frac{k_i}{a_i^2 + (\sigma - \sigma_i)^2}. \quad (4.12)$$

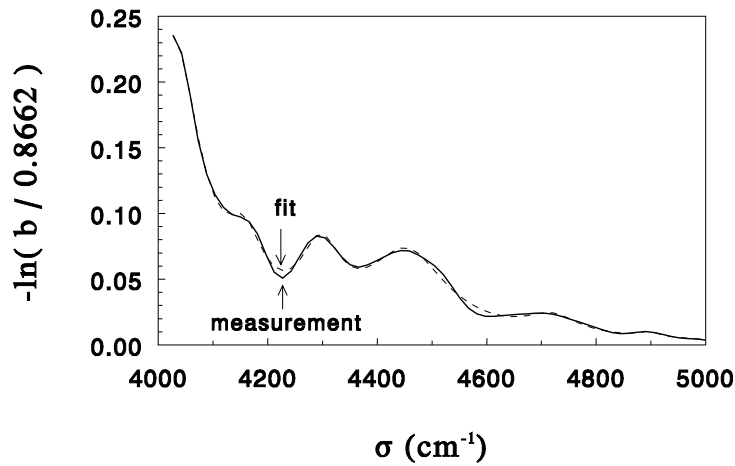


Figure 4.7: Logarithm of the measured Fourier coefficient b and matching fit of Lorentz profiles plotted versus wave number. Note the structure that relates to Lorentz profiles centered at 4024, 4158, 4296, 4454, 4719, and 4897 cm^{-1} . The Fourier coefficient was obtained by a measurement with air behind the prism. As already noted in the caption of Fig. (4.5), the value of 0.8662 is the average value of the Fourier coefficient in an undisturbed range of the spectrum from 5000 to 6000 cm^{-1} .

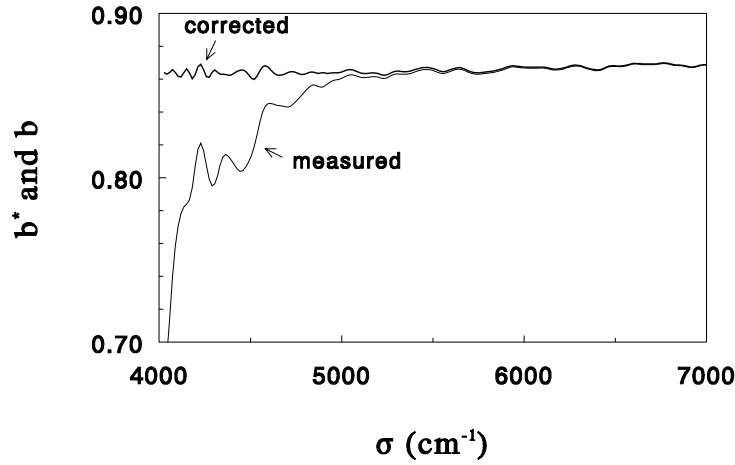


Figure 4.8: Fourier coefficient b versus wave number before (b^*) and after (b) correction for depolarization. Inaccurate alignment of polarizer and analyzer angles cause the absolute values of b to be too low: 0.8662 instead of the calculated value 0.8765. Translating these values to the refractive index of air would give n values slightly below 1. However, this does not affect the polarization degree.

A choice was made for summation of six Lorentz profiles, each centered at a particular wave number, because this best approaches the observed structure in the real part of the refractive index. The parameter k_i relates to the strength, a_i to the width, and σ_i to the central wave number of a specific absorption.

Combining Eqs. (4.10)–(4.12) yields

$$\Pi = \exp \left(-2\pi r \xi \sum_{i=1}^6 \frac{k_i \sigma}{a_i^2 + (\sigma - \sigma_i)^2} \right). \quad (4.13)$$

This relation between the polarization degree and the wave number permits the fitting of the structure due to scattering observed in the measured Fourier coefficient. In order to eliminate possible effects of the liquid behind the prism during this procedure, use is made of a measurement with air behind the BK-7 prism. The logarithm of Eq. (4.13) was fitted to the logarithm of the Fourier coefficient b and Figure (4.7) shows the results of the measurement and the fit.

With the polarization degree as a function of wave number established, measurements of liquids can be corrected for depolarization. This is done by dividing the measured Fourier coefficients a^* and b^* by the polarization degree according to Eq. (4.9). Figure (4.8) shows the results of the correction procedure. Inaccurate zero positioning of the polarizer and analyzer angles during this measurement cause the absolute values of the b -coefficients to be too low. Therefore translation of these results to refractive index values also gives values that are slightly below 1. Since only the structure seen in this measurement is used to fit to Eq. (4.13), the absolute value plays no role, and misalignment of polarizer and analyzer does

not affect the correction procedure. It should be noted that the misalignment of polarizer and analyzer occurred only during this measurement, since it was meant to be an introductory measurement that later proved to contain significant information.

4.4.2 Refractive indices

Figures (4.9)–(4.14) show the refractive indices versus wave number for the real and imaginary parts of water, Freon, and heptane and only the real parts of benzene, gas oil, and crude oil. The reason for this is that the real part of the refractive index of the latter three is so close to that of the prism that there no longer exists a situation of attenuated total reflection. This has two consequences. As most of the intensity is transmitted into the liquid, the intensity at the detector decreases dramatically. Furthermore, Δ is close to either 0° or 180° . Since it is not Δ but the cosine of Δ that is determined by rotating analyzer ellipsometry

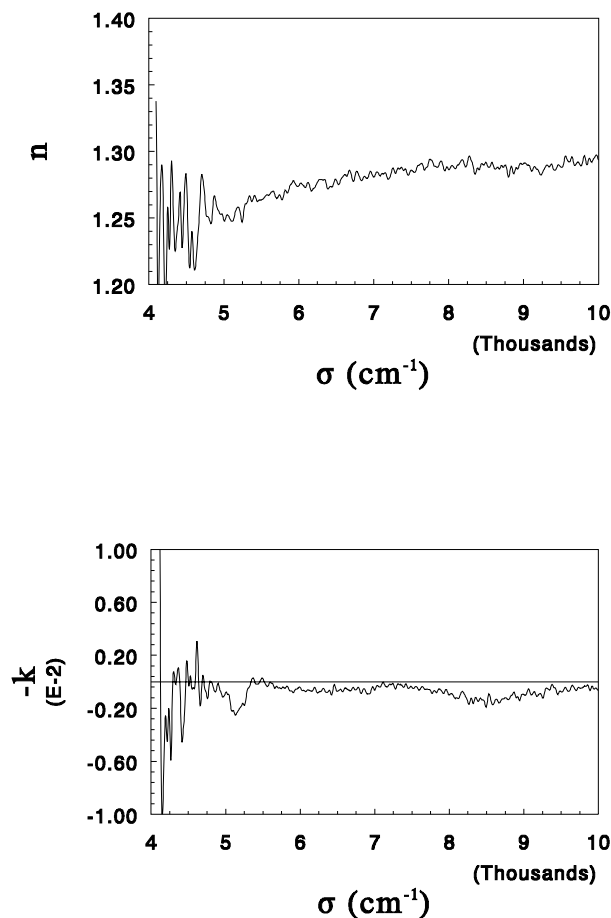


Figure 4.9: Real and imaginary parts of the refractive index of demineralized water plotted versus wave number. The structure in the imaginary part is in agreement with absorption measurements of water.

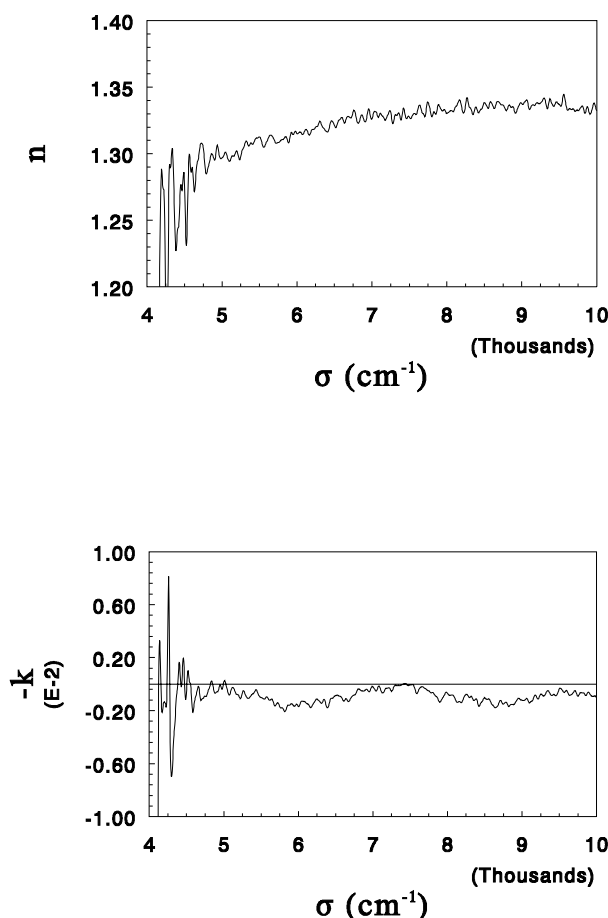


Figure 4.10: Real and imaginary parts of the refractive index of heptane $[\text{CH}_3(\text{CH}_2)_5\text{CH}_3]$ plotted versus wave number.

it is obvious that in the vicinity of $\Delta = 0^\circ$ and $\Delta = 180^\circ$ the accuracy is poor. In this case only Ψ is accurately measured, and thus only the real part, which relates to Ψ , is determined.

To demonstrate the correct working of the correction procedure, values below 5000 cm^{-1} are also shown. The intensity on the detector in this region has dropped dramatically as can be seen from the increased noise level. The average value in this region, though, gives an idea of the refractive index inclination.

The accuracy of the refractive index values is directly connected to the accuracy in Ψ and Δ , which is determined by the angle of incidence and the exact positions of the polarizer and analyzer. The angle of incidence is known to within an accuracy of $1/60^\circ$. The positions of polarizer and analyzer are known to within an accuracy of 0.01° . This translates to an accuracy of 0.001 in the refractive index values. The reproducibility or precision is largely determined by the stability of the light source on a time scale of minutes (time of measurement). Although ellipsometry is independent of the light intensity, a change in intensity

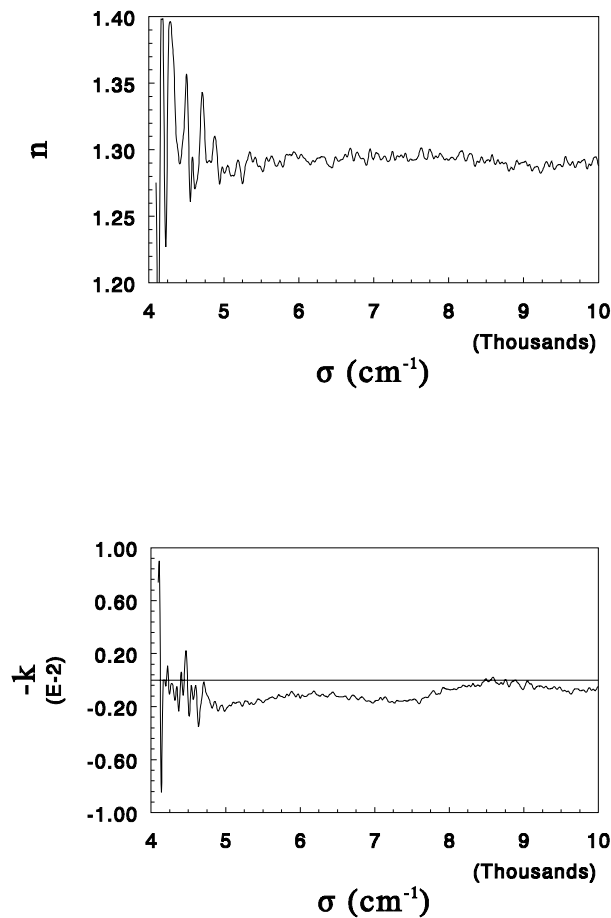


Figure 4.11: Real and imaginary parts of the refractive index of Freon 113 plotted versus wave number.

of an unstable light source within the time necessary to complete a revolution of the rotating component, causes a deformation of the sine shape of the measured intensity. This translates to deviating values of the Fourier coefficients, and subsequently calculated refractive index values. As can be seen from the graphs the precision is about 0.005 for n and 0.01 for k . Note that this implies that the relative precision is much better for n than for k .

4.5 Conclusion

Spectroscopic ellipsometry is a powerful tool for determining the complex refractive index of a liquid. Especially in combination with a prism, which effectively makes it into an ATR method, many advantages are gained over more traditional methods of refractive index determination. One advantage is that both the real and the imaginary parts are measured simultaneously, but a more important advantage is that it permits refractive index determination even for strongly ab-

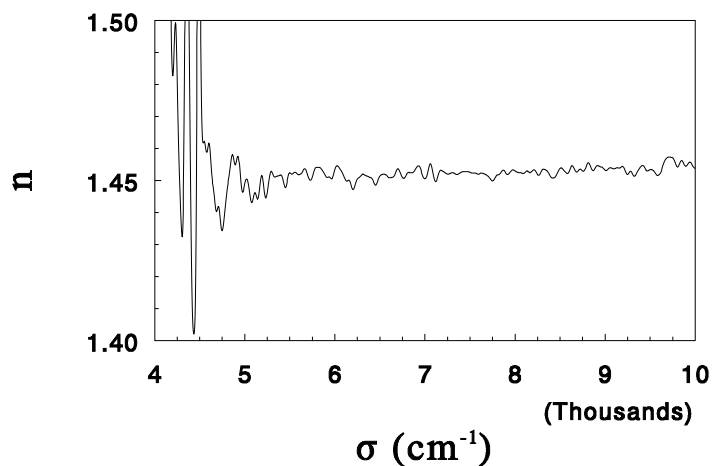


Figure 4.12: Real part of the refractive index of gas oil plotted versus wave number.

sorbing liquids like crude oil. In this situation it is the only available method capable of measuring the complex refractive index. However, for liquids that are not or only slightly absorbing the traditional specialized methods [1–7] may give more accurate results.

The results for the real part of the refractive index for water are close to what is expected in this region of the infrared, although few accurate data are available in this spectral region. The spectroscopic features in the imaginary part are in good agreement with the values obtained in absorption measurements. For all other liquids there were no data on these optical constants available in the literature, which was the main reason Shell Research Rijswijk requested us to start this investigation.

Ellipsometry appears to be very sensitive to the depolarization and scattering

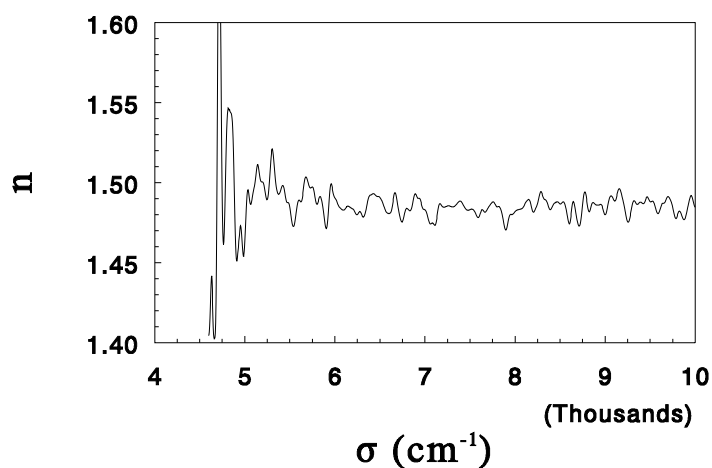


Figure 4.13: Real part of the refractive index of benzene (C_6H_6) plotted versus wave number.

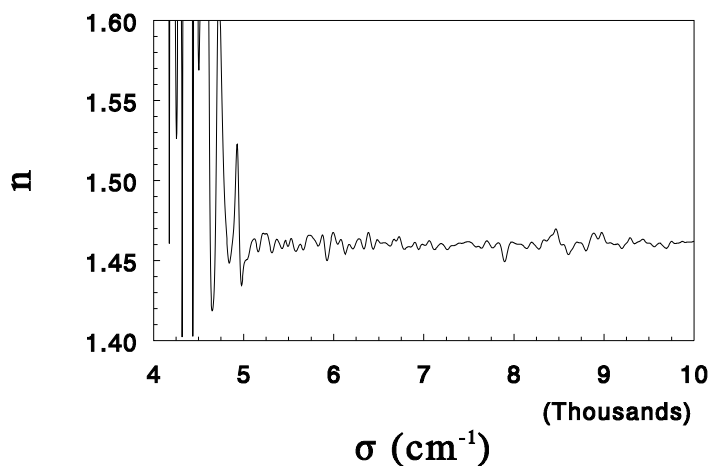


Figure 4.14: Real part of the refractive index of crude oil (Eider '89) plotted versus wave number.

that accompany absorption in the prism. Although the effects of scattering were not anticipated before the experiment, a closer study of the phenomenon yielded a very surprising new insight in the relations among absorption, scattering, and depolarization. This was expressed in a simple model which was used to correct the refractive index measurements. The model may serve as a starting point for further investigation into the rather unknown relation among absorption, scattering and depolarization.

The BK-7 prism used works well up to the cutoff wavelength, and its use may even be extended further in the infrared by application of the outlined correction procedure. However, for investigations further into the infrared or in situations where scattering is not desired, another material may be better suited for use as a prism. KRS-5 for instance is transparent up to wavelengths far in the middle infrared and will thus not scatter till far in the middle infrared. At the same time the higher refractive index (about 2.4 in the infrared) of KRS-5 ensures the working of the prism under ATR conditions even for liquids with a high refractive index.

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Chapter 5

Spectroscopic rotating compensator ellipsometry

5.1 Introduction

As was already indicated in Table (2.1) rotating compensator ellipsometry offers several distinct advantages compared to other ellipsometry methods. Among these advantages are

- The non ambiguous determination of Δ . As is clear from Eq. (3.13) it is the cosine of Δ that is determined in rotating analyzer ellipsometry. From this cosine alone it is impossible to determine the sign of Δ . As will be shown in this chapter, rotating compensator ellipsometry does not suffer this disadvantage, since it determines the tangent of Δ instead of the cosine.
- The insensitivity to source and detector polarization. In rotating compensator ellipsometry the rotating compensator is placed between two fixed polarizers. These polarizers cancel out a possible partial polarization of the source or polarization sensitivity of the detector.
- There is no need for a DC level. Rotating compensator ellipsometry determines 5 Fourier coefficients. Only three of these are necessary to determine Ψ and Δ . The DC level, represented by the 0-order Fourier coefficient, may be ignored. This is particularly interesting in cases where no good DC amplifier is available. Instead, an AC amplifier may be used.
- Complete determination of all Stokes parameters. In Section (3.2) it was argued that the determination of the circular polarization Stokes parameter S_3 requires a circular polarizer. The compensator and the analyzer in the rotating compensator setup effectively constitute such a circular polarizer. Therefore rotating compensator ellipsometry even enables determination of the S_3 Stokes parameter. Of course it also determines the other three Stokes parameters.

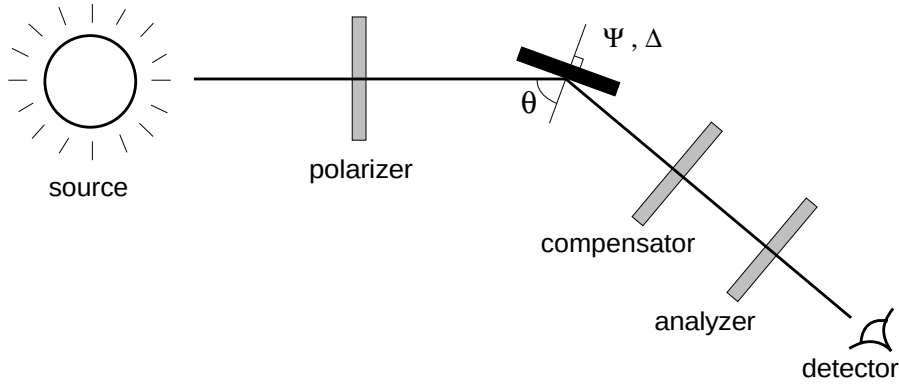


Figure 5.1: The PSCA rotating compensator configuration. The light emitted by the source traverses the polarizer, reflects at the sample, traverses the compensator and polarizer, and hits the detector

- A rotating compensator ellipsometer is self calibrating. Small imperfections of polarizer and analyzer can be translated to an imperfect compensator. The characteristics of this imperfect compensator may be determined by assuming perfect polarizers, and performing a straight through measurement. These characteristics can then be applied to ordinary rotating compensator measurements to account for the imperfections in the ellipsometer.

Despite these important advantages, one principal fact prevented a widespread application of rotating compensator ellipsometry for spectroscopic purposes. This fact is the absence of a good spectroscopic retarder. In this chapter a spectroscopic retarder will be presented, that has a retardance close to 90° over a wide range of the used spectrum. Furthermore, a sample measurement that shows the actual application of such a rotating compensator ellipsometer will be presented. But first there will be an introduction to rotating compensator ellipsometry.

5.2 Rotating compensator ellipsometer behavior

In this section the behavior of a rotating compensator ellipsometer will be calculated for a PSCA configuration. As indicated in Fig. (5.1) and reflected by the acronym, the light on entering the ellipsometer passes the polarizer, sample, compensator, and analyzer, consecutively. An alternative approach to rotating compensator ellipsometry offers the PCSA configuration, in which the compensator is positioned in front of the sample. At the end of this section the adaptations necessary for a PCSA setup will be indicated.

To describe the PSCA configuration in the Mueller formalism, the matrices from Table (3.1) are used. In analogy with the approach in Section (3.5) for

a perfect rotating analyzer setup, the source emits unpolarized light. This is described by a Stokes vector $\underline{\mathbf{S}}_0 = (1, 0, 0, 0)$. Subsequently, the light passes the following optical components

- A perfect polarizer with azimuth P (Eq. (3.6)).
- The sample under investigation.
- An imperfect retarder with a fast axis azimuth C , relative amplitude change Ψ_c , and retardance Δ_c . The differences between a perfect and an imperfect retarder lie in the retardance, which may differ substantially from the ideal value of 90° , and the attenuating properties. I.e., the amplitude of the linear polarization state parallel to the fast axis is changed by a different amount than the amplitude of the linear polarization state parallel to the slow axis. The matrix in Table (3.1) at position 4, representing this component is taken from the work of Hauge *et al.* [1]. Note that their definition of the relative amplitude change Ψ_c differs from the usual definition: $\Psi = 90^\circ - \Psi_c$.
- A perfect analyzer with azimuth A (Eq. (3.7)).

Multiplication of this train of Mueller matrices with the initial Stokes vector yields as a resulting Stokes vector

$$\underline{\mathbf{S}}_u = \underline{\mathbf{M}}_A \cdot \underline{\mathbf{R}}(A - C) \cdot \underline{\mathbf{M}}_{\Psi_c \Delta_c} \cdot \underline{\mathbf{R}}(C) \cdot \underline{\mathbf{M}}_{\Psi \Delta} \cdot \underline{\mathbf{R}}(-P) \cdot \underline{\mathbf{M}}_P \cdot \underline{\mathbf{S}}_0. \quad (5.1)$$

The same argument as used in Section (3.5) justifies the absence of the rotation matrices $\underline{\mathbf{R}}(P)$ and $\underline{\mathbf{R}}(A)$ just behind the source and in front of the detector, respectively.

Evaluation of the first component of this Stokes vector — the intensity — shows harmonic contributions of second and fourth order in the compensator azimuth angle C . Therefore the intensity may be written as

$$I(C) = A_0 + A_2 \cos 2C + B_2 \sin 2C + A_4 \cos 4C + B_4 \sin 4C, \quad (5.2)$$

where the Fourier coefficients A_0 – B_4 are given by

$$\begin{aligned}
A_0 &= \frac{1}{\pi} \int_0^\pi I(C) \, dC, \\
A_2 &= \frac{2}{\pi} \int_0^\pi I(C) \cos 2C \, dC, \\
B_2 &= \frac{2}{\pi} \int_0^\pi I(C) \sin 2C \, dC, \\
A_4 &= \frac{2}{\pi} \int_0^\pi I(C) \cos 4C \, dC, \\
B_4 &= \frac{2}{\pi} \int_0^\pi I(C) \sin 4C \, dC.
\end{aligned} \tag{5.3}$$

Further calculation of these Fourier coefficients results in

$$\begin{aligned}
A_0 &= 1 - \cos 2P \cos 2\Psi \\
&\quad + \frac{1}{2}(1 + y_c) \left[\cos 2A (\cos 2P - \cos 2\Psi) \right. \\
&\quad \left. + \sin 2A \sin 2P \sin 2\Psi \cos \Delta \right],
\end{aligned} \tag{5.4}$$

$$\begin{aligned}
A_2 &= x_c (\cos 2P - \cos 2\Psi) \\
&\quad + x_c \cos 2A (1 - \cos 2P \cos 2\Psi) - z_c \sin 2A \sin 2P \sin 2\Psi \sin \Delta,
\end{aligned} \tag{5.5}$$

$$\begin{aligned}
B_2 &= x_c \sin 2P \sin 2\Psi \cos \Delta \\
&\quad + z_c \cos 2A \sin 2P \sin 2\Psi \sin \Delta + x_c \sin 2A (1 - \cos 2P \cos 2\Psi),
\end{aligned} \tag{5.6}$$

$$A_4 = \frac{1}{2}(1 - y_c) \left[\cos 2A (\cos 2P - \cos 2\Psi) - \sin 2A \sin 2P \sin 2\Psi \cos \Delta \right], \tag{5.7}$$

$$B_4 = \frac{1}{2}(1 - y_c) \left[\cos 2A \sin 2P \sin 2\Psi \cos \Delta + \sin 2A (\cos 2P - \cos 2\Psi) \right]. \tag{5.8}$$

Here the quantities x_c , y_c , and z_c represent the non ideal behavior of the compensator

$$\begin{aligned}
x_c &\equiv \cos 2\Psi_c \approx 0, \\
y_c &\equiv \sin 2\Psi_c \cos \Delta_c \approx 0, \\
z_c &\equiv \sin 2\Psi_c \sin \Delta_c \approx 1.
\end{aligned} \tag{5.9}$$

The approximations are valid for an almost ideal compensator, i.e., $\Psi_c \approx 45^\circ$ and $\Delta_c \approx 90^\circ$. Now consider a common case in rotating compensator ellipsometry, in

which the polarizer azimuth P is 45° and the analyzer is in the plane of incidence ($A = 0^\circ$). In this situation Eqs. (5.4)–(5.8) simplify considerably

$$\begin{aligned} A_0 &= 1 - \frac{1}{2}(1 + y_c) \cos 2\Psi, \\ A_2 &= x_c (1 - \cos 2\Psi), \\ B_2 &= x_c \sin 2\Psi \cos \Delta + z_c \sin 2\Psi \sin \Delta, \\ A_4 &= -\frac{1}{2}(1 - y_c) \cos 2\Psi, \\ B_4 &= \frac{1}{2}(1 - y_c) \sin 2\Psi \cos \Delta. \end{aligned} \quad (5.10)$$

From these expressions Ψ and Δ may be solved. To that end observe three facts

1. For an almost ideal compensator the quantity x_c is close to zero. As a consequence the A_2 Fourier coefficient will also be close to zero. Application of this coefficient in solving for the ellipsometric angles will therefore introduce large errors.
2. Fourier coefficient A_0 — the DC component of the intensity — may contain interference from external light sources. Although this is not true for ellipsometry in combination with a Fourier transform spectrometer, it is better to avoid its use.
3. Since no absolute intensity is measured in ellipsometry, the absolute values of the Fourier coefficients can not be determined. Consequently, only the ratio of the Fourier coefficients may be used.

On these grounds the use of Fourier coefficients A_0 and A_2 will be dispensed with, and the other coefficients will be scaled to A_4 . This results in two expressions relating the Fourier coefficients and the compensator imperfections to the ellipsometric quantities

$$\begin{aligned} \frac{B_2}{A_4} &= -\frac{2x_c}{1 - y_c} \tan 2\Psi \cos \Delta - \frac{2z_c}{1 - y_c} \tan 2\Psi \sin \Delta, \\ \frac{B_4}{A_4} &= -\tan 2\Psi \cos \Delta. \end{aligned} \quad (5.11)$$

To simplify the resulting expressions for the ellipsometric angles, two intermediate quantities X_1 and X_2 are introduced

$$\begin{aligned} X_1 &\equiv \tan 2\Psi \sin \Delta = \frac{x_c}{z_c} \frac{B_4}{A_4} - \frac{1 - y_c}{2z_c} \frac{B_2}{A_4}, \\ X_2 &\equiv \tan 2\Psi \cos \Delta = -\frac{B_4}{A_4}. \end{aligned} \quad (5.12)$$

With the aid of these intermediate quantities the ellipsometric angles can be expressed as

Table 5.1: To determine the values of Ψ and Δ , the signs of Fourier coefficients A_4 and B_4 are required. The sub-indices used in conjunction with Ψ and Δ indicate the step in the procedure, e.g., 1 indicates the original value as obtained from Eq. (5.13), 2 indicates the sign after considering the sign of the Fourier coefficient, etc..

$A_4 > 0 \implies \Psi_2 := 90^\circ - \Psi_1$ $A_4 < 0 \implies \Psi_2 := \Psi_1$	
$B_4 > 0 \implies \Delta_2 := \Delta_1$ $B_4 < 0 \implies \Delta_2 := \Delta_1 + 180^\circ$	$\Delta_2 < 0 \implies \Delta_3 := \Delta_2 + 360^\circ$ $\Delta_2 > 0 \implies \Delta_3 := \Delta_2$

$$\begin{aligned}\tan 2\Psi &= \sqrt{X_1^2 + X_2^2}, \\ \tan \Delta &= X_1/X_2.\end{aligned}\tag{5.13}$$

Since the value of Ψ is determined from a root and the arctan function returns only values between -90° and $+90^\circ$, the exact values of Ψ and Δ have to be established using sign information from the Fourier coefficients. If, for example, the A_4 coefficient is positive, it follows from Eq. (5.10) that $\cos 2\Psi$ is negative and, subsequently, $\tan 2\Psi$ is also negative since $\sin 2\Psi$ is always positive. This implies that the negative root of Eq. (5.13) has to be taken or, equivalently, the value of Ψ already obtained has to be negated. Finally, to bring Ψ into the correct range, a period of $\tan 2\Psi - 90^\circ$ — has to be added. The complete procedure is outlined in Table (5.1).

Eq. (5.13) clearly shows that the tangents of both Ψ and Δ are determined by rotating compensator ellipsometry. This implies that there are no regions in the Ψ – Δ plane where the ellipsometer dramatically loses accuracy.

Similar relations are easily derived for other azimuths of polarizer and analyzer. To obtain the relations for PCSA configuration the azimuths P and A in Eqs. (5.4)–(5.8) must be exchanged. Subsequently, the procedure outlined for PSCA must be followed.

5.3 Design and characterization of a spectroscopic retarder

The retarders commonly used in single wavelength applications are quarter wave plates. It is obvious that, since these plates are designed for one optimum wavelength, they cannot be used for spectroscopic purposes. Furthermore, in the infrared the same argument as used for the absence of good polarizers is valid: most birefringent materials become opaque for wavelengths greater than $5\ \mu\text{m}$.

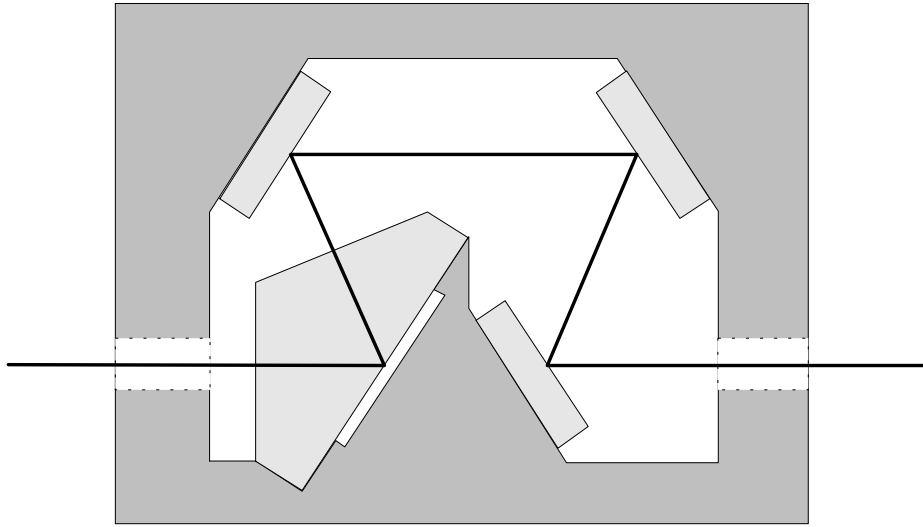


Figure 5.2: The spectroscopic retarder containing a KRS-5 prism retarder and three gold coated mirrors. The prism is slightly asymmetric to prevent multiple reflections. The prism face on which the light beam is incident makes an angle of 33.2° and the plane through which the beam leaves makes an angle of 34.2° with the prism base. All components are mounted on an aluminum frame.

Thus the development of a spectroscopic IR retarder requires a concept which is completely different from the simple birefringent retarder as used in single wavelength applications in the visible region of the spectrum.

5.3.1 Design of a retarder

The concept for a spectroscopic retarder presented here is that of total internal reflection. This well known phenomenon has since long been applied in retarders such as the Fresnel and Mooney rhombs [2]. Application in ellipsometry [3], however, was hampered by the disturbed collinearity of the incident and reflected light beams. This would be acceptable for a static retarder, such as used in rotating analyzer ellipsometry to bring Δ in a more favorable region, but obviously not for rotating compensator ellipsometry. Besides this principal design criterion, several other demands were imposed on the design

- The retarder should be compact so it can be mounted on a rotation table.
- The retarder should exhibit the least possible dispersion.
- The retarder should contain a minimum number of mirrors.
- The retarder should be able to deal with beam diameters up to 1 cm.

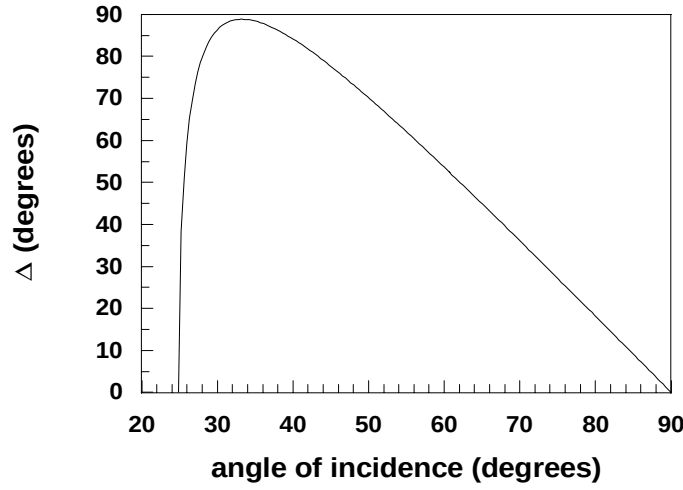


Figure 5.3: The retardance generated by a KRS-5 prism as a function of the angle of incidence. At the critical angle the retardance starts to rise from 0 and it reaches its maximum value Δ_m at angle θ_m .

The actual design sketched in Fig. (5.2) reflects these demands. The KRS-5 prism, the three gold coated mirrors, and their respective positions in the aluminum frame all affect the polarizing behavior of the retarder. The following sections deal with this behavior.

KRS-5 prism

The prism is positioned such that a beam collinear with the rotation axis of the retarder, enters the prism perpendicularly. Subsequently, the light reflects at the prism base at an angle of incidence of 33.2° . The aluminum retarder frame leaves an air gap at the prism base and since the angle of incidence is larger than the critical angle of 24.90° , total internal reflection occurs at the KRS-5–air interface. For light with a frequency of 1250 cm^{-1} ($n_{\text{KRS-5}} = 2.375$), the reflection introduces a phase difference of 88.65° between the p - and s -polarization states. This is calculated using Eq. (4.4) [4], which governs the phase shift occurring at total internal reflection. Observe in Fig. (5.3) that for the chosen angle of incidence the generated phase difference reaches its maximum value. The maximum value, Δ_m , and the angle for which it occurs, θ_m , depend only on the refractive index values and are given by

$$\tan \frac{1}{2} \Delta_m = \frac{n_0^2 - n_1^2}{2n_0 n_1}, \quad \sin^2 \theta_m = \frac{2n_1^2}{n_0^2 + n_1^2}, \quad (5.14)$$

where n_0 and n_1 represent the refractive indices of KRS-5 and air, respectively.

After the light beam reflects at the prism base, the beam exits the prism at an oblique angle. In this way the occurrence of multiple reflections inside the prism is prevented. The small angle of incidence, however, introduces a

Table 5.2: For each gold coated mirror the angle of incidence, the relative amplitude change Ψ and the retardance Δ are given. The beam is assumed to enter the retarder collinear with the rotation axis and to have a frequency of 1250 cm^{-1} . At this frequency the refractive index of gold is $8.5 - j46.4$.

Mirror	θ_0 ($^\circ$)	Ψ ($^\circ$)	Δ ($^\circ$)
1	32.47	44.93	179.18
2	33.20	44.92	179.14
3	33.20	44.92	179.14

relative amplitude change of the two mutually perpendicular polarization states. In common with the ATR prism in Section (4.2.3), this situation is described by the Fresnel transmission coefficients in Eq. (4.6) leading to a relative amplitude change given by Eq. (4.7). Again, the phase remains unchanged.

Gold mirrors

To make the light beam collinear with the rotation axis of the retarder, three gold coated mirrors are used. These mirrors each introduce a phase shift and a relative amplitude change given by Eq. (2.2) and dependent solely on the angle of incidence and the refractive indices of air and gold. The first mirror is positioned such that after reflection the beam becomes parallel with the rotation axis of the retarder. The other two mirrors displace the beam back to the rotation axis. Table (5.2) presents the angles of incidence on the three mirrors and the generated phase shifts and amplitude changes. Note that allowances had to be made for the finite dimension of the cutting tool that produced the retarder frame. This implies that the mirrors cannot be placed exactly in a corner of the frame. Note, furthermore, that for other frequencies the angles of incidence change. As a result of dispersion in KRS-5 each frequency leaves the prism with a different angle of refraction.

Table 5.3: Total polarizing behavior of the retarder with separate contributions of the prism internal reflection, prism to air transmission, mirror 1 reflection, and mirror 2 reflection. The contribution of the reflection at mirror 3 is equal to the one of mirror 2. Besides the resulting Ψ and Δ , also the angle of incidence, θ_0 is given. All calculations are made for a frequency of 1250 cm^{-1} .

	reflection prism	transmission prism	reflection mirror 1	reflection mirror 2	total
θ_0 ($^\circ$)	33.19	1.04	32.47	33.20	x
Ψ ($^\circ$)	45.00	45.01	44.93	44.92	44.77
Δ ($^\circ$)	88.65	0.00	179.18	179.14	-93.88

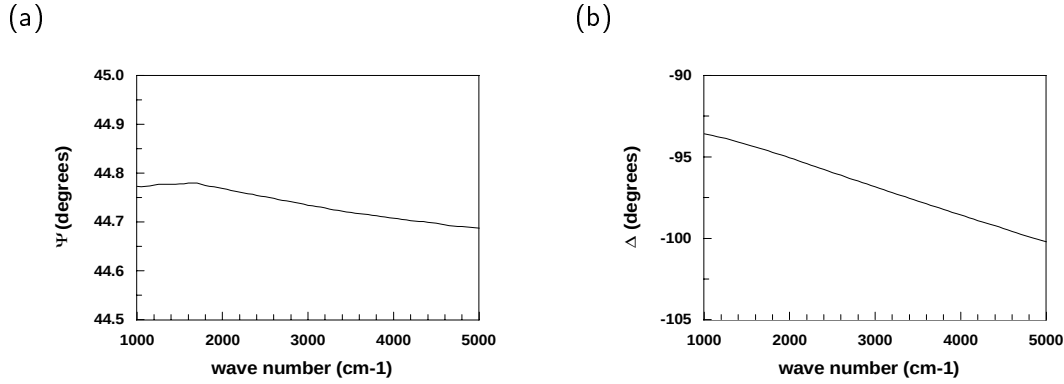


Figure 5.4: Polarizing properties of the complete retarder as a function of wave number. (a) Ψ versus wave number. (b) Δ versus wave number. The necessary dispersion functions for gold and KRS-5 can be obtained from references [5] and [6], respectively.

Complete retarder

To completely describe the polarizing properties of the retarder all polarizing contributions mentioned in the previous sections have to be collected. Observe that from the definition of Ψ and Δ in Eq. (2.2) it follows that the tangents of Ψ have to be multiplied, whereas the retardances Δ have to be added. The separate contributions and the resulting polarizing properties for the complete retarder at 1250 cm⁻¹ are presented in Table (5.3). The results as functions of wave number are shown in Fig. (5.4).

Observe in Table (5.3) the minus sign for the total retardance. This implies that the s -component is ahead of the p -component, i.e., the s -direction defines the fast axis of the retarder. That means that the fast axis stands at a right angle to the frame of the retarder, i.e., perpendicular to the plane of drawing in Fig. (5.2).

5.3.2 Characterization of a retarder

To perform reliable measurements with a rotating compensator, it is imperative to exactly know the polarizing properties of the retarder in use. These polarizing properties are obtained from a measurement without a sample in the ellipsometer in the so called straight through measurement mode. In this situation Eq. (5.10) can be simplified by inserting $\Psi = 45^\circ$ and $\Delta = 0^\circ$

$$\begin{aligned}
A_0 &= 1, \\
A_2 &= x_c, \\
B_2 &= x_c, \\
A_4 &= 0, \\
B_4 &= \frac{1}{2}(1 - y_c).
\end{aligned} \tag{5.15}$$

To keep the characterization procedure more general, it will be assumed that the azimuth of the retarder is unknown. In this case the Fourier coefficients will depend on the zero position of the compensator azimuth relative to the polarizer azimuth, C , and will transform according to Eq. (5.31)

$$\begin{aligned}
A_0 &= 1, \\
A_2 &= x_c(\cos 2C + \sin 2C), \\
B_2 &= x_c(-\sin 2C + \cos 2C), \\
A_4 &= \frac{1}{2}(1 - y_c) \sin 4C, \\
B_4 &= \frac{1}{2}(1 - y_c) \cos 4C.
\end{aligned} \tag{5.16}$$

Solving this for the compensator characteristics yields

$$\begin{aligned}
x_c &= \pm \frac{\sqrt{2A_2^2 + 2B_2^2}}{2A_0}, \\
y_c &= 1 - 2 \frac{\sqrt{A_4^2 + B_4^2}}{A_0}.
\end{aligned} \tag{5.17}$$

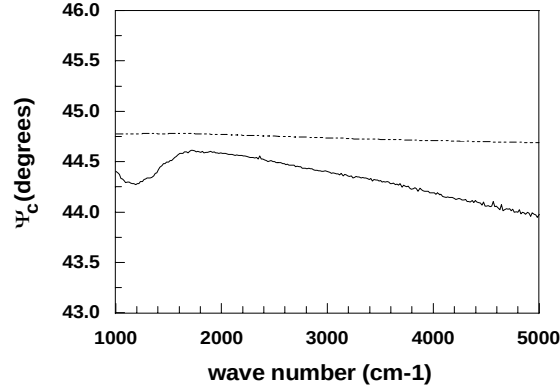
The sign of x_c cannot be determined with this method without further knowledge of the retarder. For the retarder under investigation, though, the approximate position of the fast axis is known. Moreover, from the simulation in the previous section it is known that Ψ_c is smaller than 45° . More to this effect will be said towards the end of this section. When the characterization measurement starts with the fast axis close to the zero azimuth and Ψ_c is known to be smaller than 45° , the plus sign in Eq. (5.17) applies.

The values of Ψ_c and Δ_c can be derived from Eq. (5.9)

$$\begin{aligned}
\cos 2\Psi_c &= x_c, \\
\cos \Delta_c &= \frac{y_c}{1 - x_c^2}.
\end{aligned} \tag{5.18}$$

Despite the resemblance of these expressions to those for rotating analyzer ellipsometry in Eq. (3.13), they do not suffer the problems usually encountered in rotating analyzer ellipsometry. The reasons are that for a good retarder with a retardance close to 90° , the arccos-function is well behaved and that the sign of Δ_c is positive by definition, since the fast axis is close to the zero azimuth.

(a)



(b)

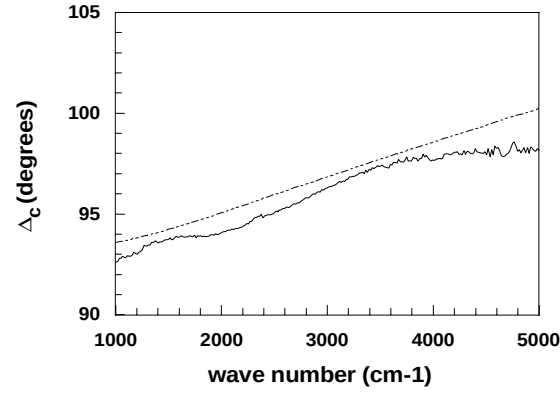


Figure 5.5: Polarizing properties of the complete retarder as a function of wave number. Solid lines indicate measured data, dashed lines calculated data. (a) Ψ_c versus wave number. (b) Δ_c versus wave number.

The difference in sign for $\cos 2\Psi_c$ between Eqs. (3.13) and (5.18) arises from the definition of Ψ_c in Table (3.1): $\Psi_c = 90^\circ - \Psi$.

Beside the values of x_c and y_c , the zero azimuth of the compensator relative to the polarizer may also be found from Eq. (5.16) and is given by

$$\tan 2C = \frac{A_2 - B_2}{A_2 + B_2} \quad \text{or} \quad \tan 4C = \frac{A_4}{B_4}. \quad (5.19)$$

Note that these two expressions can be used to obtain separate compensator azimuths for the second and fourth harmonic contributions to the intensity in Eq. (5.2). This is useful when amplification circuits in the intensity detection system exhibit a frequency dependence. In a rotating compensator setup in which the retarder is moved in steps, this does not occur, but in a setup with a continuously rotating retarder, it may.

The actual measurement of the retarder characteristics is performed with a setup similar to the one shown in Fig. (3.1). The Rochon polarizers are replaced with tandem wire grids and with the retarder between them. The retarder is

positioned with the fast axis close to the transmission axis of the first polarizer. The transmission axis of the second polarizer is put at an azimuth of 45° relative to the transmission axis of the first one. With this starting position a standard rotating compensator measurement is performed, i.e., the compensator is stepped and for each step a spectrum is measured. From a cross-section through these spectra at a fixed wave number the Fourier coefficients are determined, and hence the values of Ψ_c and Δ_c . Results and expected values are presented in Fig. (5.5). It should be noted that because the measurement starts with the fast axis close to the zero azimuth, the measured Δ_c is positive and should be equal to the negated values of Δ in Fig. (5.4b). Furthermore, $\tan \Psi_c$ represents the ratio of the amplitude changes along the slow and the fast axes. As already noted the slow axis coincides with the p -direction of the prism and the fast axis with the s -direction. Therefore the values of Ψ_c and Ψ in Fig. (5.4a) should be equal.

The deviations between the measured and the expected values of Ψ_c and Δ_c , though small, seem to be systematic. One possible source of error may be the imperfection of the wire grid polarizers. This is supported by the deviations that increase with wave number. As mentioned in the introduction of this chapter, the fact that the measured retarder properties also comprise the imperfections in the rest of the setup is actually a strong advantage of rotating compensator ellipsometry.

Other error sources may be beam wander due to misalignment of the retarder to the incident beam, depolarization in the prism [3], and misalignment of the components in the setup. Exact determination of the origin of the deviations requires further experiment and simulation.

Finally some words to the construction of the retarder. Best results are achieved in an approach in which the prism is manufactured before the frame. In that case the exact dimensions of the prism may be determined. For custom made optical components these sometimes differ considerably from the dimensions specified to the manufacturer. When the prism dimensions are known, these can be inserted in a ray tracing program to obtain the exact specifications for the frame. In this way the frame is tailored to the prism. The gold or possibly aluminum mirrors should not be coated. These coatings, used to prevent damage or increase reflectance, cause the polarizing properties of the mirrors to change dramatically, and disrupt the correct operation of the retarder.

5.4 Calibration of the rotating compensator ellipsometer

The next step in the preparation of the ellipsometer after the determination of the compensator characteristics, is the calibration of the azimuth angles of all components relative to the plane of incidence. The procedure here is very similar to the one for a rotating analyzer ellipsometer in Section (3.6). The compensator is removed from the setup and the remaining rotating analyzer setup is calibrated.

Once the polarizer and analyzer azimuths are determined, they are put at 0° and 90° for the polarizer and analyzer, respectively. Another option is to put both the polarizer and the analyzer at 0° relative to the plane of incidence. If now the compensator is moved back in the setup and it is assumed that the fast axis is exactly in the plane of incidence, Eqs. (5.4)–(5.8) with appropriate values for P and A apply

$$\begin{aligned} A_0 &= \tfrac{1}{2}(1 - y_c)(1 - \cos 2\Psi), \\ A_2 &= 0, \\ B_2 &= 0, \\ A_4 &= -\tfrac{1}{2}(1 - y_c)(1 - \cos 2\Psi), \\ B_4 &= 0. \end{aligned} \tag{5.20}$$

Of course the position of the fast axis is unknown at the time of calibration and therefore a compensator azimuth error dC has to be taken into account. This causes the Fourier coefficients to be transformed according to Eq. (5.31)

$$\begin{aligned} A_4 &= -\tfrac{1}{2}(1 - y_c)(1 - \cos 2\Psi) \cos 4dC, \\ B_4 &= \tfrac{1}{2}(1 - y_c)(1 - \cos 2\Psi) \sin 4dC. \end{aligned} \tag{5.21}$$

Only the fourth harmonic Fourier coefficients are shown because the others either vanish or remain unchanged. The compensator azimuth error dC is defined in a fashion similar to the polarizer and analyzer azimuth errors in Eq. (3.14)

$$C \equiv C' + dC, \tag{5.22}$$

where C is the real azimuth of the fast axis and C' the azimuth as it is read from the rotation table. The error angle is now straightforwardly calculated from Eq. (5.21)

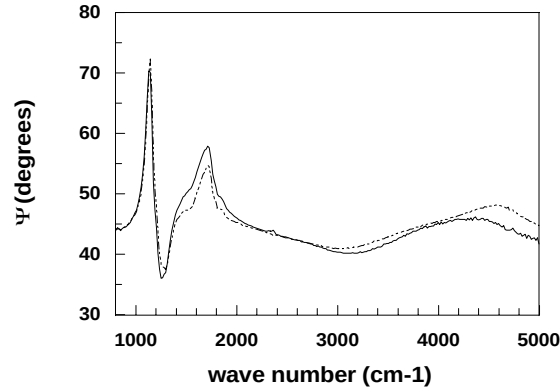
$$\tan 4dC = -\frac{B_4}{A_4}. \tag{5.23}$$

Now all azimuth angles and the compensator characteristics are known, the ellipsometer is ready for measurements.

5.5 Comparison between the rotating analyzer and compensator ellipsometers

To establish the correct operation of the rotating compensator ellipsometer, the results obtained from a measurement at a calibration sample similar to the one discussed in Section (3.6), are compared with a rotating analyzer measurement at the same sample. The rotating analyzer results are calculated from the relevant Fourier coefficients assuming that the setup is perfect, i.e., the expressions in

(a)



(b)

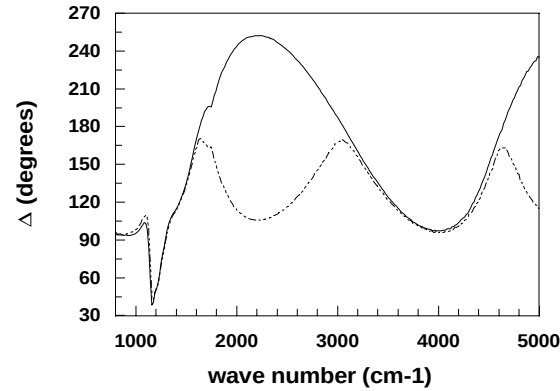


Figure 5.6: (a) Ψ and (b) Δ as measured with a rotating compensator ellipsometer and a rotating analyzer ellipsometer versus wave number. Solid lines indicate rotating compensator, dashed lines indicate rotating analyzer results.

Eq. (3.13) are used. The results for the rotating compensator are calculated with the expressions in Eq. (5.13) and the compensator characteristics in Fig. (5.5). The values of Ψ and Δ as measured with the two ellipsometers are presented in Fig. (5.6).

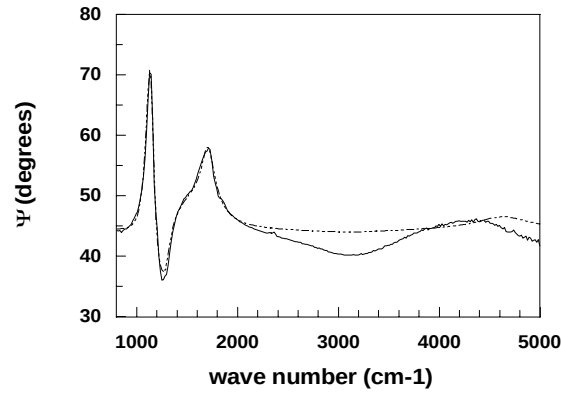
The most striking feature in Fig. (5.6) is the presence of values of Δ obtained with rotating compensator ellipsometry in excess of 180° . This clearly reveals the capacity to measure Δ in the range from 0 to 360° , whereas the results for Δ obtained with rotating analyzer ellipsometry are obviously limited to values below 180° . It is also obvious from Fig. (5.6) that in the three wave number regions where Δ approaches 180° , the values obtained with rotating analyzer ellipsometry become unreliable.

The differences between the rotating compensator and analyzer results can be attributed to the neglect of imperfections in the rotating analyzer setup and errors in the calibration of the compensator, as discussed in the previous section.

5.6 Application to $(\text{CF}_x)_n$ -layer

Finally, the rotating compensator ellipsometer is applied to measure the characteristics of a teflon like layer. The sample under investigation — the calibration sample from the previous section — is prepared starting with a silicon wafer. On this wafer an aluminum layer is deposited under the following plasma conditions: at a pressure of 10 mTorr and a flow of 100 sccm Argon, a plasma is burnt with a power of approximately 80 W during 15 minutes. The atoms sputtered from an aluminum target are deposited on the silicon wafer. The expected thickness of the aluminum layer is approximately 200 nm. Since this thickness amply exceeds the skin depth, the aluminum may be regarded as the substrate. On this substrate a teflon like layer $((\text{CF}_x)_n)$ is deposited with a typical refractive index of about

(a)



(b)

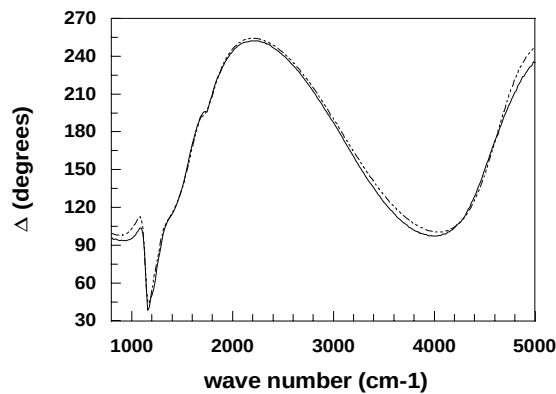


Figure 5.7: (a) Ψ and (b) Δ as measured with a rotating compensator ellipsometer and their respective fits. The fitting model consists of an aluminum substrate, a layer with a certain thickness and two harmonic oscillators contributing to a certain background refractive index. The ambient is air. The angle of incidence is 68° . Solid lines indicate measurements, dashed lines indicate fit results.

Table 5.4: Characteristics of the harmonic oscillators in the teflon layer. The background refractive index is $1.474 - j 0.0011$. The thickness is $1.4 \mu\text{m}$.

central wave number (cm^{-1})	width (FWHM) (cm^{-1})	strength
1201	153	0.1888
1707	157	0.0098

1.4. The teflon layer is deposited with a 100 Watts CHF_3 plasma at 300 mTorr and a flow of 25 sccm during 15 minutes.

The results of the ellipsometric measurements at an angle of incidence of 68° are shown in Fig. (5.7). In this figure also the results from a fit to the measurement data are presented. The model that is fitted consists of an aluminum substrate, a layer with a certain thickness and background refractive index with contributions from 2 harmonic oscillators. The ambient is air. Results of the fit of this model to the measurement are given in Table (5.4).

The background refractive index and layer thickness derived from the fit are in agreement with expectations. The oscillation at 1201 cm^{-1} is readily identified with C-F stretch bonds [7] in the layer. For the oscillation at 1707 cm^{-1} , though, there exist three candidates. The C=C stretch bond [8], normally at 1600 cm^{-1} , may have shifted towards 1700 cm^{-1} under influence of surrounding bonds. Another possibility is the compound bond of H-C-F. However, the best candidate is the C=O stretch bond at 1700 cm^{-1} [9]. In that case one would have to suppose a leak in the vacuum system to supply the necessary oxygen.

The measurement and fit presented in Fig. (5.7) show good agreement except for Ψ at wave numbers of 2000 cm^{-1} and higher. The discrepancies may have been caused by the thickness of the aluminum layer. If a thickness of 10 nm is assumed instead of the expected 200 nm, the aluminum layer can no longer be considered as the substrate, and simulations show the same features as the measurement in the vicinity of 2500 cm^{-1} . Another, more probable cause is the already mentioned leak in the vacuum system. In that case the aluminum layer may have oxidized during deposition. This would lower the refractive index of the aluminum layer and make it partly transparent.

Note that in the rotating analyzer results on the same sample presented in the previous section, similar features in the vicinity of 2500 cm^{-1} are present. This makes the explanation of the discrepancy by a measurement error unlikely.

5.7 Conclusion

In this chapter the method of rotating compensator ellipsometry was presented as the superior way of performing ellipsometry. Only when extreme demands are made on the speed, phase modulation ellipsometry should be preferred. The difficulty in spectroscopic rotating compensator ellipsometry, however, was the lack

of a good retarder with a well defined and more or less wavelength independent retardance. To solve this problem the design of a spectroscopic retarder was put forward, based on total internal reflection in a KRS-5 prism. The expected polarizing properties of this retarder were calculated and subsequently compared with experimentally obtained data. Although the agreement of the calculated and the measured properties looks promising, the small systematic differences between the two call for further study. As a test case the rotating compensator ellipsometer results were compared with rotating analyzer results. Besides a clear display of the advantage of rotating compensator ellipsometry in the determination of Δ , again good agreement between the results could be established. Finally, the rotating compensator ellipsometer was used to measure the characteristics of a teflon like layer on aluminum. Measurement and fit show good agreement except for some parts of the spectrum. Oxidization of the aluminum substrate is the most probable cause.

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Appendix

Error propagation in a rotating compensator ellipsometer

To calculate the effects of errors in the azimuth angles or component imperfections on the resulting ellipsometric quantities measured with a rotating compensator ellipsometer, an approach similar to the one of De Nijs [10, 11] for a rotating analyzer ellipsometer is chosen. In this approach the errors in Ψ and Δ are related to the error sources using perturbation matrices. Since not Ψ and Δ but the Fourier coefficients A_0 – B_4 are directly measured, the first step in the error determination is the relation between the errors in Ψ and Δ on the one hand and the errors in the Fourier coefficients on the other hand. Then only the relations between the errors in for example the polarizer azimuth and the errors in the Fourier coefficients have to be established.

The material presented here may serve as a starting point for more extensive sensitivity calculations, expressing the errors in Ψ and Δ also in the polarizer and analyzer angles.

Errors in the Fourier coefficients

The easiest way to express the errors of Ψ and Δ in the Fourier coefficient errors is to start with Eq. (5.11). From this expression the Jacobian matrix, relating the errors in Ψ and Δ to those in the scaled Fourier coefficients B_2/A_4 and B_4/A_4 , may be derived

$$\begin{bmatrix} \frac{\partial \Psi}{\partial (B_2/A_4)} & \frac{\partial \Psi}{\partial (B_4/A_4)} \\ \frac{\partial \Delta}{\partial (B_2/A_4)} & \frac{\partial \Delta}{\partial (B_4/A_4)} \end{bmatrix} = \frac{\cos 2\Psi}{4z_c} \begin{bmatrix} -\cos 2\Psi \sin \Delta (1 - y_c) & 2 \cos 2\Psi (x_c \sin \Delta - z_c \cos \Delta) \\ -2 \frac{\cos \Delta (1 - y_c)}{\sin 2\Psi} & 4 \frac{x_c \cos \Delta + z_c \sin \Delta}{\sin 2\Psi} \end{bmatrix}. \quad (5.24)$$

To arrive at this expression, the partial derivatives of Eq. (5.11) with respect to Ψ and Δ are calculated. Subsequently, the resulting matrix is inverted. Although Eq. (5.24) may be used to calculate the errors in Ψ and Δ from the errors in the scaled Fourier coefficients, it is easier to calculate them from the errors in the unscaled Fourier coefficients in Eq. (5.10). Therefore Eq. (5.24) has to be multiplied by the Jacobian matrix relating the errors in the unscaled Fourier coefficients to those in the scaled ones

$$\begin{bmatrix} \frac{\partial(B_2/A_4)}{\partial B_2} & \frac{\partial(B_2/A_4)}{\partial A_4} & \frac{\partial(B_2/A_4)}{\partial B_4} \\ \frac{\partial(B_4/A_4)}{\partial B_2} & \frac{\partial(B_4/A_4)}{\partial A_4} & \frac{\partial(B_4/A_4)}{\partial B_4} \end{bmatrix} = \frac{2}{(1-y_c)\cos 2\Psi} \begin{bmatrix} -1 & 2\tan 2\Psi \frac{x_c \cos \Delta + z_c \sin \Delta}{1-y_c} & 0 \\ 0 & -\tan 2\Psi \cos \Delta & -1 \end{bmatrix}. \quad (5.25)$$

Multiplication of Jacobian matrices (5.24) and (5.25) yields

$$\begin{bmatrix} \frac{\partial \Psi}{\partial B_2} & \frac{\partial \Psi}{\partial A_4} & \frac{\partial \Psi}{\partial B_4} \\ \frac{\partial \Delta}{\partial B_2} & \frac{\partial \Delta}{\partial A_4} & \frac{\partial \Delta}{\partial B_4} \end{bmatrix} = \begin{bmatrix} \frac{\cos 2\Psi \sin \Delta}{2z_c} & \frac{\sin 2\Psi}{1-y_c} & -\frac{\cos 2\Psi}{z_c(1-y_c)}(x_c \sin \Delta - z_c \cos \Delta) \\ \frac{\cos \Delta}{z_c \sin 2\Psi} & 0 & -2\frac{x_c \cos \Delta + z_c \sin \Delta}{z_c(1-y_c) \sin 2\Psi} \end{bmatrix}. \quad (5.26)$$

Perturbation calculus in ellipsometry

As De Nijs already showed, the first order errors in one of the components in the ellipsometer setup can be related to a perturbed Stokes vector. This is achieved by replacing the Mueller matrix of the relevant component in Eq. (5.1) with its perturbing counterpart, i.e., its first order derivative with respect to the perturbing quantity. The Fourier coefficients, obtained from integration — a linear operation — of the first component of the Stokes vector, are then linear in the perturbing error.

Polarizer and analyzer azimuth

Consider the polarizer azimuth P for example. The perturbing rotation Mueller matrix is given by

$$\frac{\partial \underline{\mathbf{R}}(-P)}{\partial P} = 2 \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -\sin 2P & -\cos 2P & 0 \\ 0 & \cos 2P & -\sin 2P & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (5.27)$$

Substituting this perturbing matrix into Eq. (5.1) yields for the perturbed Stokes vector

$$\frac{\partial \underline{\mathbf{S}}_u}{\partial P} = \underline{\mathbf{M}}_A \cdot \underline{\mathbf{R}}(A-C) \cdot \underline{\mathbf{M}}_{\Psi_c \Delta_c} \cdot \underline{\mathbf{R}}(C) \cdot \underline{\mathbf{M}}_{\Psi \Delta} \cdot \frac{\partial \underline{\mathbf{R}}(-P)}{\partial P} \cdot \underline{\mathbf{M}}_P \cdot \underline{\mathbf{S}}_0. \quad (5.28)$$

Complete evaluation of this expression and subsequent Fourier analysis for $P = 45^\circ$ and $A = 0^\circ$ finally gives the perturbed Fourier coefficients

$$\begin{pmatrix} \frac{\partial B_2}{\partial P} \\ \frac{\partial A_4}{\partial P} \\ \frac{\partial B_4}{\partial P} \end{pmatrix} = \begin{pmatrix} 0 \\ -(1 - y_c) \\ 0 \end{pmatrix}. \quad (5.29)$$

Note that this vector can also be obtained directly by taking the derivatives of Eqs. (5.6)–(5.8) with respect to the polarizer azimuth and subsequent substitution of $P = 45^\circ$ and $A = 0^\circ$. Multiplication of vector (5.29) with the Jacobian (5.26) results in the desired relation between the errors in the ellipsometric quantities and the polarizer azimuth error

$$\begin{pmatrix} \frac{\partial \Psi}{\partial P} \\ \frac{\partial \Delta}{\partial P} \end{pmatrix} = \begin{pmatrix} -\sin 2\Psi \\ 0 \end{pmatrix}. \quad (5.30)$$

A similar procedure can be followed to obtain the perturbed Ψ and Δ values as a function of the analyzer azimuth error. Results are presented in Table (5.5).

Compensator azimuth

A special case is the calculation of the errors caused by the compensator azimuth C . The integration of the intensity over this angle has caused this quantity to drop out, and therefore the errors in the ellipsometric angles cannot be computed directly by taking the derivatives of the Fourier coefficients with respect to the angle C . To deal with the compensator azimuth thus requires another approach. Looking at Eq. (5.2), it is easy to see that if one chooses another origin for the compensator azimuth, the Fourier coefficients transform as follows

$$\begin{pmatrix} A'_0 \\ A'_2 \\ B'_2 \\ A'_4 \\ B'_4 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \cos 2C & \sin 2C & 0 & 0 \\ 0 & -\sin 2C & \cos 2C & 0 & 0 \\ 0 & 0 & 0 & \cos 4C & \sin 4C \\ 0 & 0 & 0 & -\sin 4C & \cos 4C \end{bmatrix} \begin{pmatrix} A_0 \\ A_2 \\ B_2 \\ A_4 \\ B_4 \end{pmatrix}, \quad (5.31)$$

where the primes indicate transformed quantities and C now represents the difference between the original azimuth origin and the new one. The differential

Table 5.5: First order errors in the relevant Fourier coefficients — B_2 , A_4 , and B_4 — and Ψ and Δ , caused by azimuth errors and polarizer imperfections.

		$\begin{pmatrix} B_2 \\ A_4 \\ B_4 \end{pmatrix}$	$\begin{pmatrix} \Psi \\ \Delta \end{pmatrix}$
Polarizer azimuth	$\frac{\partial}{\partial P}$	$\begin{pmatrix} 0 \\ -(1-y_c) \\ 0 \end{pmatrix}$	$\begin{pmatrix} -\sin 2\Psi \\ 0 \end{pmatrix}$
Analyzer azimuth	$\frac{\partial}{\partial A}$	$\begin{pmatrix} 2x_c \\ -\sin 2\Psi \cos \Delta (1-y_c) \\ -\cos 2\Psi (1-y_c) \end{pmatrix}$	$\frac{1}{z_c} \begin{pmatrix} -x_c \cos 2\Psi (1+\cos 2\Psi) \sin \Delta + z_c \cos \Delta \\ 2x_c (1+\cos 2\Psi) \cos \Delta + 2z_c \cos 2\Psi \sin \Delta \end{pmatrix}$
Compensator azimuth	$\frac{\partial}{\partial C}$	$\begin{pmatrix} -2x_c (1-\cos 2\Psi) \\ 2(1-y_c) \sin 2\Psi \cos \Delta \\ 2(1-y_c) \cos 2\Psi \end{pmatrix}$	$\frac{-1}{z_c} \begin{pmatrix} x_c \cos 2\Psi (1+\cos 2\Psi) \sin \Delta - 2z_c \cos \Delta \\ \frac{2x_c (1+\cos 2\Psi) \cos \Delta + 4z_c \cos 2\Psi \sin \Delta}{\sin 2\Psi} \end{pmatrix}$
Polarizer imperfection	$\frac{\partial}{\partial \alpha_P}$	$\begin{pmatrix} -\sin 2\Psi (x_c \cos \Delta + z_c \sin \Delta) \\ -\frac{1}{2} \cos 2\Psi (1-y_c) \\ -\frac{1}{2} \sin 2\Psi \cos \Delta (1-y_c) \end{pmatrix}$	$\begin{pmatrix} -\cos 2\Psi \sin 2\Psi \\ 0 \end{pmatrix}$
Analyzer imperfection	$\frac{\partial}{\partial \alpha_A}$	$\begin{pmatrix} \sin 2\Psi (x_c \cos \Delta - z_c \sin \Delta) \\ \frac{1}{2} \cos 2\Psi (1-y_c) \\ -\frac{1}{2} \sin 2\Psi \cos \Delta (1-y_c) \end{pmatrix}$	$\begin{pmatrix} \frac{x_c}{z_c} \cos 2\Psi \sin 2\Psi \cos \Delta \sin \Delta \\ 2 \frac{x_c}{z_c} \cos^2 \Delta \end{pmatrix}$

change in the Fourier coefficients is now equal to

$$\begin{pmatrix} \frac{\partial A_0}{\partial C} \\ \frac{\partial A_2}{\partial C} \\ \frac{\partial B_2}{\partial C} \\ \frac{\partial A_4}{\partial C} \\ \frac{\partial B_4}{\partial C} \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -2 \sin 2C & 2 \cos 2C & 0 & 0 \\ 0 & -2 \cos 2C & -2 \sin 2C & 0 & 0 \\ 0 & 0 & 0 & -4 \sin 4C & 4 \cos 4C \\ 0 & 0 & 0 & -4 \cos 4C & -4 \sin 4C \end{bmatrix}_{C=0} \begin{pmatrix} A_0 \\ A_2 \\ B_2 \\ A_4 \\ B_4 \end{pmatrix}, \quad (5.32)$$

which evaluates for the relevant Fourier coefficients to

$$\begin{pmatrix} \frac{\partial B_2}{\partial C} \\ \frac{\partial A_4}{\partial C} \\ \frac{\partial B_4}{\partial C} \end{pmatrix} = \begin{pmatrix} -2A_2 \\ 4B_4 \\ -4A_4 \end{pmatrix} = \begin{pmatrix} -2x_c (1 - \cos 2\Psi) \\ 2(1 - y_c) \sin 2\Psi \cos \Delta \\ 2(1 - y_c) \cos 2\Psi \end{pmatrix}. \quad (5.33)$$

Multiplication of Jacobian (5.26) with this vector finally results in the relation between errors in the ellipsometric quantities and the compensator azimuth

$$\begin{pmatrix} \frac{\partial \Psi}{\partial C} \\ \frac{\partial \Delta}{\partial C} \end{pmatrix} = \frac{-1}{z_c} \begin{pmatrix} x_c \cos 2\Psi(1 + \cos 2\Psi) \sin \Delta - 2z_c \cos \Delta \\ \frac{2x_c(1 + \cos 2\Psi) \cos \Delta + 4z_c \cos 2\Psi \sin \Delta}{\sin 2\Psi} \end{pmatrix}. \quad (5.34)$$

Polarizer and analyzer imperfection

Consider the partial polarizer in Table (3.1), position 2. Then its perturbing counterpart is given by

$$\frac{\partial \underline{\underline{\mathbf{M}}}}{\partial \alpha} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{\alpha} & 0 \\ 0 & 0 & 0 & 1/\sqrt{\alpha} \end{bmatrix}. \quad (5.35)$$

Substitution of this matrix into Eq. (5.1) at the position of the polarizer and the analyzer yields the relations between the errors in the Fourier coefficients and the ones in the attenuation coefficients of the polarizer (α_P) and the analyzer (α_A), respectively. From these errors and the Jacobian matrix (5.26) the errors in Ψ and Δ may be calculated. Results are presented in Table (5.5).

Chapter 6

Utilization

Right from the initial stages of this STW project — the development of a spectroscopic rotating analyzer ellipsometer — interest was shown in the instrument itself and its possibilities. The Bruker company, as the manufacturer of the spectrometer, was interested in the spectroscopic ellipsometer as an optional extension to their spectrometer. Their participation in the project was therefore a logical step. During the actual development more interest was generated, which led to several projects in cooperation with Shell Research Amsterdam, Shell Research Rijswijk, and Océ Research. In the following sections the aims, methods, and results of these projects will be discussed briefly. This serves more as an example of the application of spectroscopic ellipsometry in other fields of chemistry and physics than a specific scientific purpose.

6.1 Interface layer between polymer films

Shell Research Amsterdam (KSLA) was interested in determining the thickness and composition of the interface layer between two layers of partly miscible polymer systems. This interface layer comes into existence as a result of annealing at temperatures above the glass transition temperature. The development in time of the interface layer was of particular interest to KSLA. After a study to ascertain the feasibility of spectroscopic ellipsometry in the determination of small changes in interface layer, a program was carried out in which at regular time intervals a polymer sample was taken from an oven and its ellipsometric spectra were measured. The small changes in the ellipsometric spectra were translated to the desired information on diffusion. The polymer samples of main interest, polymethyl methacrylate (PMMA) and polystyrene-co-acrylonitrile (SAN), were manufactured by spin coating polymer solutions onto a gold coated silicon wafer. Typical ellipsometric results obtained for a polymer layer on gold are shown in Fig. (6.1). From these measurements it was concluded that the ellipsometer yielded interesting qualitative results but needed further refinement to gain better quantitative information on diffusion in the interface layer. The conversion of ro-

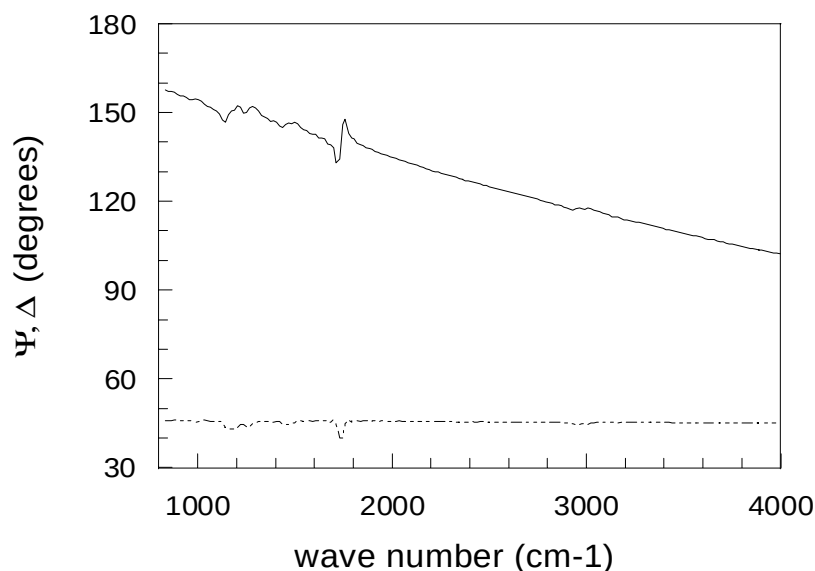


Figure 6.1: Typical rotating analyzer results for a PMMA layer on a gold coated silicon wafer as measured for KSLA. Ψ is indicated by a dashed line, Δ by a solid line. The structure observed in both Ψ and Δ between 1000 cm⁻¹ and 2000 cm⁻¹ is characteristic for PMMA.

tating analyzer to rotating compensator ellipsometry, in particular, was regarded as a way to obtain better results.

6.2 The complex refractive index of several liquids in the infrared

The interest from Shell Research Rijswijk (KSEPL) in spectroscopic infrared ellipsometry was aroused by the necessity to know the complex refractive index of several liquids in the infrared. Especially for the strongly absorbing liquids the traditional methods could not measure the real part of the refractive index. The values of the complex refractive index were necessary to verify the correct operation of a new instrument to detect small polluting concentrations of for example oil in waste water before it is drained off. The instrument is based on absorption and scattering of infrared light. The method of attenuated total reflection used in this utilization project and the obtained results are presented in Chapter 4 of this thesis. Besides the refractive indices of the liquids given there, the refractive indices of several other liquids were determined, such as sea water, production water from several drilling sites, 1-heptene, toluene, silicon oil, and paraffine.

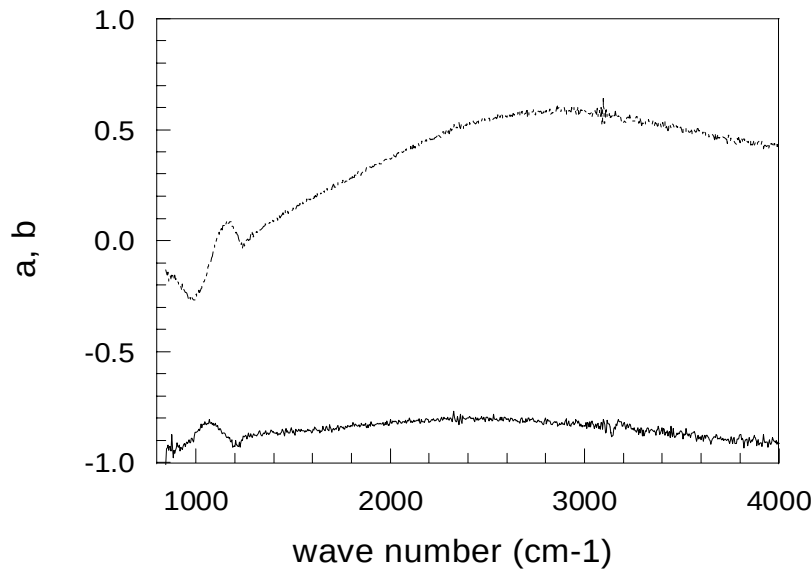


Figure 6.2: Typical rotating analyzer results as measured for Océ for a silicon wafer with a sandwich of SiO_x layers. In this case the Fourier coefficients a (solid line) and b (dashed line) are shown instead of Ψ and Δ .

6.3 Interface between silicon oxide layers

The involvement of Océ Research in the project resulted from the wish to characterize the interface between layers of silicon oxides (SiO_x) where the fraction of oxygen is different for each layer. Results of this research find their application in the determination of wear in drums used in copiers. Since the bonds between silicon and oxygen exhibit a pronounced activity in the infrared, spectroscopic infrared ellipsometry seemed an appropriate diagnostic tool. On this basis a series of test measurements were performed on samples prepared by Océ and it was at that time clear that ellipsometry showed great promise, but still needed a lot of work. In particular the translation of the Ψ and Δ values to the desired quantities and the unfavorable regions in the Δ -plane close to 0 and 180° required attention. Results of a measurement on a sample prepared by Océ are shown in Fig. (6.2).

6.4 Further development of the spectroscopic ellipsometer

The observations made by Océ formed the basis of a new project. In this project, in which the Eindhoven University of Technology (TUE), the Technology Foundation (STW), and Bruker participate, the spectroscopic ellipsometer is prepared for use in applied projects. To achieve this the rotating analyzer ellipsometer

principle was replaced with the superior rotating compensator principle. The implications of this step are reflected in Chapter 5 of this thesis. Furthermore, the outdated computer system for the control of the spectrometer was exchanged for a personal computer. This permitted the creation of a user friendly interface to the measurement procedures of the spectroscopic ellipsometer and, very importantly, the development of a simulation-fit program that helps in the interpretation of the measurement results. The main features of this program are presented in Chapter 2 of this thesis.

Chapter 7

Conclusions and recommendations

7.1 Summary of conclusions

One of the main problems in spectroscopic infrared ellipsometry was the absence of good polarizers. The idea of tandem wire grids, evaluated in this thesis, helps to solve this problem. The tandem wire grid performs quadratically better than a single wire grid. A single wire grid polarizer introduces a phase difference of 90° between the component along the transmission axis and the component perpendicular to this axis.

The measured characteristics of polarizers, source, and detector can be used to correct the results obtained in rotating analyzer ellipsometry for component imperfections.

With the aid of the Mueller matrix description for all components in the setup, the behavior of the setup is calculated and subsequently a calibration procedure is derived. The calibration procedure requires a sample with a value for Ψ close to 45° and a value for Δ which varies substantially as a function of wavelength. A metal sample with a thick transparent layer on top satisfies these conditions. The calibration procedure is based on the fact that the variations in the residue, caused by the variations in Δ , vanish for a polarizer position in the plane of incidence.

Ellipsometry is able to measure the real and imaginary parts of the refractive index of a material simultaneously. Moreover, in combination with an attenuated total reflection prism applied to strongly absorbing liquids, it is the only method capable of measuring the complex refractive index. However, for liquids that are not or only slightly absorbing the traditional specialized methods may give more accurate results.

The results for the real part of the refractive index for water are close to what is expected in this region of the infrared, although few accurate data are available in this spectral region. The spectroscopic features in the imaginary part are in good agreement with the values obtained in absorption measurements.

Ellipsometry appears to be very sensitive to the depolarization and scattering that accompany absorption in the prism. Although the effects of scattering were not anticipated before the experiment, a closer study of the phenomenon yielded a very surprising new insight into the relation among absorption, scattering, and depolarization. This was expressed in a simple model which was used to correct the refractive index measurements. The model may serve as a starting point for further investigation into the rather poorly known relation among absorption, scattering and depolarization.

The BK-7 prism used works well up to the cutoff wavelength, and its use may even be extended further into the infrared by application of the outlined correction procedure. However, for investigations further into the infrared or in situations where scattering can not be tolerated, another material may be better suited for use as a prism. KRS-5, for instance, is transparent up to wavelengths far in the middle infrared and will thus not scatter till far in the middle infrared. At the same time the higher refractive index (about 2.4 in the infrared) of KRS-5 ensures the working of the prism under ATR conditions even for liquids with a high refractive index.

A spectroscopic infrared retarder based on total internal reflection in a KRS-5 prism is presented. The expected polarizing properties of this retarder were calculated and subsequently compared with experimentally obtained data. Although the agreement of the calculated and the measured properties looks promising, the small systematic differences between the two call for further study.

The retarder developed in the present work is essential in the realization of a spectroscopic infrared rotating compensator ellipsometer. As a test case for this instrument the results were compared with rotating analyzer results. Besides a clear display of the advantage of rotating compensator ellipsometry in the determination of Δ , again good agreement between the results could be established. Finally, the rotating compensator ellipsometer was used to measure the characteristics of a teflon like layer on aluminum. Measurement and fit show good agreement except for some parts of the spectrum. Oxidization of the aluminum substrate is the most probable cause.

7.2 Recommendations

Integrated tandem wire grid polarizer

A disadvantage of the current tandem wire grids, especially of the ones with a KRS-5 substrate ($n = 2.4$), is loss of intensity caused by reflection. The extra wire grid introduces two extra interfaces, air–KRS-5 and KRS-5–air, that each pass only 83% of the incident intensity. Although the situation for BaF₂ substrates ($n = 1.5$) is not as dramatic, it is evident that the extra polarizer reduces the intensity. The loss of intensity can be prevented if the two wire grids are mounted on both sides of the substrate. Other advantages of this method are that for once and for all the two grids can be accurately aligned by the manufacturer and that the total polarizer becomes more compact.

Refractive indices of liquids

As was shown in Chapter 4, the imaginary parts of the refractive indices of gas oil, benzene, and crude oil could not be determined because the refractive index of the liquid was too close to that of the BK-7 prism. Application of KRS-5 as prism material may solve this and at the same time prevent the depolarization and scattering as a consequence of absorption in the BK-7 glass. Sensitivity may be gained by designing the prism angles such that an angle of incidence of just above the critical angle occurs. It can be understood from Fig. (5.3) that Δ is very sensitive to changes in the refractive index for angles of incidence just above the critical angle.

Improving the retarder

The main source of dispersion in the retarder is the reflection at the three gold coated mirrors. Therefore the simplest way to reduce the dispersion is to decrease the number of mirrors or decrease the angles of incidence on these mirrors. The number of mirrors, however, cannot be decreased without substantially increasing the angles of incidence on the remaining mirrors, whereas the angles of incidence can only be decreased at the cost of extra mirrors or scaling up of the entire retarder frame. Another option is the application of different mirror materials, e.g., the Δ and the dispersion in Δ generated by aluminum coated mirrors will be smaller than that by gold coated mirrors. A very interesting alternative option is to slightly change the angle of incidence towards smaller angles. A first consequence is that the dispersion in the prism increases slightly. Since the contribution of the dispersion in the prism is in a direction opposite to the ones of the contributions of the mirrors, the total dispersion decreases. A second effect is that the angles of incidence on the mirrors can be reduced. This decreases the Δ generated by the mirrors and also the contributed dispersion.

A completely new approach towards the retarder design is to fully implement it in one prism. If four equal angles of incidence and a generated retardance of

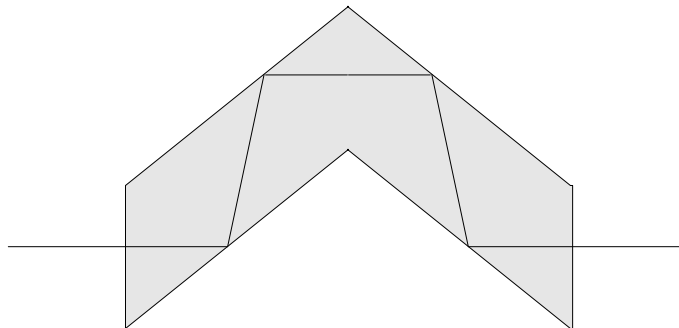


Figure 7.1: Proposal for a new prism. The prism material is KRS-5. All angles of incidence are 51.6° , each generating a retardance of 67.5° .

270° are assumed, the necessary angle of incidence can be solved from Eq. (4.4). For KRS-5 at 1250 cm^{-1} this would result in angles of incidence of 26.5° or 51.6° . The latter of these two angles should be preferred, because as is obvious from Fig. (5.3), Δ is less sensitive to errors in the angles and dispersion. A major advantage of this design is that dispersion does not cause spatial beam splitting. A drawback is the more complicated manufacturing of the prism. The proposed design, sketched in Fig (7.1), looks like two Fresnel prisms in tandem.

Improved calibration procedures

The most time consuming step in the preparation of the ellipsometer for measurement is the calibration procedure. For the rotating compensator ellipsometer there is an added step to this procedure in the removal of the compensator from the setup. In a turnkey system — an ellipsometer should within a few years evolve to such a system — this is unacceptable. The calibration procedure is more or less the same as finding the global extremes of a multi-dimensional function. Perhaps a smart root-finding algorithm combined with measurements at carefully selected positions of all components in the setup could appreciably speed up the calibration, without the necessity to remove the compensator from the ellipsometer.

Continuously rotating analyzer/compensator

In the current implementation of the spectroscopic ellipsometer, the rotating component is stepped. A tremendous gain in measurement speed could be gained, if the component rotates continuously, because in that case the time consuming acceleration and deceleration of the stepped component are no longer needed. The continuous rotation implies that during a measurement of the spectrum, the intensity changes due to a change in the analyzer or compensator azimuth. However, if the spectrum is measured sufficiently fast relative to the azimuth change, the induced changes in the spectrum depend only in first order on the rotating component azimuth. With the spectrometers available nowadays, scan

rates of up to 200 Hz may be attained, and implementation of this proposal is feasible. An important item in this scheme is the accurate synchronization between the scanner in the spectrometer and the stepper motors.

Integrated ATR ellipsometer

An idea of which the realization may be further into the future is the complete integration of a spectroscopic ellipsometer on an ATR prism. The light coming from the spectrometer is directed through fibers to the polarizer, ATR prism, compensator, analyzer, and from there through a fiber back to the detector. The ATR prism can then be pressed against the sample from which the refractive index, structure, etc., are to be determined. The main demand that the implementation of this concept makes is the availability of good fibers for infrared radiation. The design becomes even simpler if these fibers are capable of preserving the state of polarization. In that case the polarizers and compensator do not have to be mounted on the ATR prism but can instead be built into the spectrometer. Unfortunately, such fibers do not yet exist, although progress is being made in the development of infrared fibers. Other requirements are an already mentioned improved calibration procedure without the necessity to remove the compensator from the setup and miniaturization of the components. For the polarizers and even the compensator the miniaturization is rather straightforward. In the case of the compensator one should realize that the dimensions scale with the desired beam diameter.

Summary

Spectroscopic ellipsometry in the infrared is a promising diagnostic tool for the study of interfaces and layered systems. The development and application of a spectroscopic infrared ellipsometer as a combination of a Fourier transform spectrometer and a rotating analyzer ellipsometer is the main theme of this thesis. A major problem in the implementation of a spectroscopic ellipsometer in the infrared is the absence of good polarizing components. Therefore the description of an imperfect polarizer — a wire grid polarizer described as a partial polarizer — and the improvement of this polarizer are two important issues. Based on the description and characterization of the partial polarizer and all other imperfect components in the setup the behavior of the complete ellipsometer is calculated. This finally results in a calibration procedure for a spectroscopic ellipsometer with imperfect components. An improved polarizer in the infrared is constructed by putting two wire grid polarizers in tandem.

With the calibrated rotating analyzer ellipsometer measurements were conducted of the complex refractive index of several liquids. To warrant sufficient intensity on the detector these measurements were performed under Attenuated Total Reflection conditions. This approach, combining ATR and ellipsometry, resulted in a unique measurement principle which is capable of measuring the complex refractive index, even for strongly absorbing liquids like crude oil.

The next step in the development of the spectroscopic ellipsometer is the conversion of the rotating analyzer to the rotating compensator principle. The superior rotating compensator principle offers several distinct advantages compared to the rotating analyzer principle. Most notable are the ability to determine the ellipsometric angle Δ unambiguously and the insensitivity to source and detector polarization. An essential component in rotating compensator ellipsometry is a retarder which introduces a retardance between the two mutually perpendicular components of the electric field. In a spectroscopic implementation of rotating compensator ellipsometry this retardance should have a more or less constant value close to 90° over a wide range of the used spectrum. Thus the development of a spectroscopic infrared retarder requires a concept which is completely different from the birefringent retarder as used in single wavelength applications in the visible region of the spectrum. Therefore the successful design and characterization of the spectroscopic retarder are discussed at some length.

To establish the correct operation of the rotating compensator setup the results obtained from a measurement at a calibration sample are compared with a rotating analyzer measurement at the same sample. Comparison between the two ellipsometric principles clearly reveals the advantages of rotating compensator ellipsometry. The small differences can be attributed to the neglect of imperfections in the rotating analyzer setup. As a further test the rotating compensator ellipsometer is used to measure the characteristics of a teflon like layer on aluminum. Measurement and fit show good agreement except for some parts of the spectrum. This discrepancy is attributed to oxidization of the aluminum substrate.

Samenvatting

Spectroscopische infrarood ellipsometrie is een veelbelovende diagnostiek bij het onderzoek aan grensvlakken en gelaagde systemen. De ontwikkeling en toepassing van zo'n spectroscopische ellipsometer — de combinatie van een Fouriertransformatie spectrometer en een roterende analysator ellipsometer — is het hoofdthema in dit proefschrift. Een groot probleem bij de verwezenlijking van een spectroscopische ellipsometer voor het infrarood is het gebrek aan goede polariserende componenten. Daarom spelen de beschrijving van een imperfecte polarisator — een wire grid polarisator beschreven als een partiële polarisator — en de verbetering van deze polarisator een belangrijke rol. Uitgaande van de beschrijving en karakterisatie van de partiële polarisator en alle overige componenten in de opstelling wordt de hele ellipsometer doorgerekend. Deze aanpak resulteert in een kalibratieprocedure voor een spectroscopische ellipsometer met imperfecte componenten. Een verbeterde polarisator wordt gerealiseerd door twee wire grid polarisatoren achter elkaar te zetten.

Met de gekalibreerde opstelling werden vervolgens metingen verricht om de complexe brekingsindex van verschillende vloeistoffen te bepalen. Om voldoende intensiteit te waarborgen werd gekozen voor een ATR (Verzwakte totale reflectie) benadering. Deze benadering in combinatie met ellipsometrie levert een unieke meetmethode op die in staat is zelfs voor sterk absorberende vloeistoffen zoals ruwe olie de complexe brekingsindex te bepalen.

De volgende stap in de ontwikkeling van de spectroscopische ellipsometer werd ingezet door de overgang van het roterende analysator naar het roterende compensator principe. Roterende compensator ellipsometrie biedt enkele duidelijke voordelen ten opzichte van roterende analysator ellipsometrie. Meest opvallende voordelen zijn de mogelijkheid om het teken van de ellipsometrische hoek Δ te bepalen en de ongevoeligheid voor bron- en detectorpolarisatie. Een essentiële component hierbij is de retardator, die een faseverschil tussen de twee onderling loodrechte componenten van het elektrische veld teweegbrengt. In spectroscopische toepassingen moet dit faseverschil over het totale spectrum een nagenoeg constante waarde in de buurt van 90° hebben. Dit vereist een volledig ander werkingsprincipe dan de dubbelbrekende retardator zoals deze gebruikt wordt voor monochromatisch licht in het zichtbare spectrum. Daarom wordt veel aandacht besteed aan het ontwerp en karakterisatie van deze spectroscopische retardator.

Als test worden de resultaten verkregen met roterende compensator ellipsometrie aan een kalibratiesample vergeleken met die van roterende analysator ellipsometrie aan hetzelfde sample. Hierbij vallen direct de voordelen van roterende compensator ellipsometrie op. De kleine verschillen worden geweten aan de verwaarlozing van imperfecties in de roterende analysator ellipsometer. Tenslotte werd de roterende compensator ellipsometer ingezet om de optische eigenschappen van een teflon-achtige laag op aluminium te bestuderen. Meting en aanpassing tonen alleen verschillen in sommige delen van het spectrum. Deze verschillen worden toegeschreven aan oxidatie van het aluminium.

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